FedRAS: Network-Conditioned Rank Allocation for Federated LoRA

Wenyu Ma[a,b,c], Tao Hu[a,b,c,*], Xinlei Liu[a,b,c]

[a]*Information Engineering University, Zhengzhou 450001, China*
[b]*Key Laboratory of Cyberspace Endogenous Safety and Security of Henan Province, Zhengzhou 450001, China*
[c]*Key Laboratory of Cyberspace Security, Ministry of Education of China, Zhengzhou 450001, China*

Abstract

In mobile-edge federated learning, uplink links are time-varying and deadline-bound, and full-parameter updates frequently miss their deadlines when links degrade. Federated LoRA reduces ~~update size, but a fixed rank cannot adapt to time-varying links. We formulate~~ the payload, but existing systems treat the rank as a training hyperparameter, ignoring the coupling that the rank itself determines the payload and the payload determines the delivery probability: a larger rank buys adaptation capacity but is more likely to miss the deadline. We formalize rank selection and client admission as a joint ~~network scheduling problem and call the resulting method FedRAS. It estimates a delivery probability and an adaptation capacity for each~~ scheduling problem under deadline constraints and propose FedRAS, which scores every client–rank ~~pair, maximizes the expected useful updates under byte-budget and participation constraints~~pair with a payload-conditioned delivery risk and a saturating capacity gain, solves the assignment exactly with a dynamic program under a byte budget and a participation cap, and aggregates heterogeneous ranks with frozen-factor rank masking. ~~All payloads are accounted at their real serialized sizes: a frozen-A client uploads only the trainable B factor, 0.090 MB per round at rank 16, against 5.29 MB for a full-model update.~~

~~Experiments on non-IID CIFAR-10, CIFAR-100, and MNIST with a~~

*Corresponding author
Email address: hutaondsc@163.com (Tao Hu)

500-client scheduler simulation use deadlines recalibrated to these real payloads. At the nominal deadline , rank selection reduces to fixed rank-16 behavior at zero cost. Under a 0.1 s deadlineThe design comes with mechanism-level guarantees: the grid assignment is exact on the stated budget grid, the budget regime in which a per-client greedy rule is already optimal is characterized in closed form, and frozen-factor aggregation of heterogeneous ranks is exact. The gains come from delivering more updates within the deadline rather than from changing training, which is why accuracy remains statistically equivalent while the network-side outcomes improve. On a real trainer, FedRAS delivers 22% more timely updates and with 23% fewer bytes than fixed rank 16 over ten seeds (p=0.002), while a full-model update misses under deadline stress at statistically equivalent accuracy, while full-parameter updates miss every stress deadlineand spends 58.7× the bytes. A pre-registered TOST equivalence test confirms that accuracy is preserved. The dynamic program improves on greedy scheduling by 28 to 55% when ; link-oblivious low-rank schedules trade adaptation capacity for delivery, and FedRAS holds the high-capacity end of that trade-off; and, at 500 clients, exact scheduling beats greedy heuristics whenever the budget binds, and by up to 64% over value-per-byte pricing, at 23–421 ms solver cost, and a deliberately miscalibrated delivery predictor costs at most 1% of the assignment objective. These results describe network-side gains in the evaluated frozen-backbone setting and do not imply universal accuracy gainsat sub-second per-round cost.

Keywords: Federated LoRA, Mobile edge computing, Rank selection, Deadline scheduling, Wireless federated learning

1. Introduction

Mobile edge networks increasingly host federated learning (FL) workloadsin which heterogeneous clients such as , with clients that include smartphones, vehicles, UAV terminals, and satellite-connected devices collaborate on a shared model without exposing raw data [1–4][1, 2], spanning IoT deployments [5] and space–air–ground systems [3, 4]. The dominant cost in of such systems is the uplink. Full-parameter : full-parameter model updates saturate wireless capacity , and drain client energy [6], and frequently miss round deadlines when links degrade during handover or outage , as reported for ground, aerial, and satellite segments [7–10]. Low-rank adapta-

tion (LoRA) has therefore become an attractive transmission format~~for FL, because~~, because each adapted layer communicates only two small factor matrices ~~per adapted layer are communicated~~ instead of the full weight delta [11–13].

Existing ~~Federated~~ federated LoRA systems, however, treat the rank r as a model hyperparameter~~. The~~, whether fixed across the fleet [14, 15], varied across clients [16, 17], adapted across rounds [18, 19], or personalized through model-heterogeneous and dual-adapter designs [20, 21]. In all of these systems the rank is selected from data and system constraints alone, ~~possibly varying across clients [16, 17] or adapting across rounds [18, 19], but not~~ never from the state of a lossy, deadline-constrained link. This convention ignores a scheduling dimension that is directly visible to the network. The omitted coupling is mechanical~~. A~~: a larger rank transmits more bytes and therefore loses delivery probability exactly where it gains adaptation capacity, while a smaller rank travels light but forfeits capacity on the healthy links that could have carried it. Whether the extra capacity is worth the delivery risk depends on the link state, and no training-side rule observes that state. Each round must therefore decide jointly, under a byte budget and a participation cap, which clients participate and at what rank.

This raises three questions ~~. How should~~ that must be answered. First, how should the delivery probability depend on ~~rank and~~ the rank and the observable link state? ~~How should~~ The rank directly determines the payload size, and the payload determines whether an update can meet the deadline on a lossy link, so the delivery model must take the rank as an argument rather than an afterthought. Second, how should the client–rank assignment be solved under budget and participation constraints? ~~Does~~ The problem is a multiple-choice knapsack with an admission constraint, greedy rules are not necessarily optimal, and one must establish when exact solving is necessary and when it may degrade to a cheap rule. Third, does the resulting adaptation improve network behavior in a real trainer? ~~These questions are answered in Sections 3, 4, and 6, respectively~~The scheduling objective is a network-side utility, not the training accuracy, and the two need not agree, so the validation must show that accuracy does not degrade because of the scheduling.

1.1. ~~Contributions~~

We answer these questions with a delivery model, a grid-based dynamic-programming solver, a rank-selection rule, and a two-tier evaluation~~. We~~

~~call the~~ , and call the resulting method FedRAS. ~~The contributions are as follows~~Our contributions are threefold.

- ~~**Deadline-aware client rank model**~~**FedRAS framework.** We ~~formulate each round as a joint assignment in which~~ propose FedRAS, to our knowledge the first framework that treats the LoRA rank as a deadline-constrained, network-conditioned, schedulable resource: each round, client admission and rank are ~~selected together. The model links rank to its real serialized payload, a delivery probability $P_{\text{on}}(r)$, and a saturating capacity gain $G(r)$. Sections 3and 3.1 develop the model~~decided jointly under a byte budget and a participation cap, with payloads accounted at their real serialized sizes (Section 3).

- ~~**Dynamic-programming scheduler**~~**Theoretical analysis.** ~~**The estimated problem is a multiple-choice knapsack with an admission cap. We derive a pseudo-polynomial dynamic program that is exact for the discretized single-round estimated assignment problem, with complexity $O(|\mathcal{D}|KB_{\max}|\mathcal{R}|)$. The solver study measures greedy gaps and runtime.**~~

- ~~**Heterogeneous-rank aggregation.**~~ ~~**We use rank-masked averaging with a shared frozen A factor, so only the trainable B factor is ever uploaded. Under this condition , the rule gives exact coordinate-wise averaging of active induced-update slices.**~~ We prove exactness of the grid assignment (Proposition 1), the budget-feasibility condition under which the cheap per-client rule is already optimal (Proposition 2), and exact frozen-factor aggregation of heterogeneous ranks (Proposition 3~~states the conditions and limits~~), with bounded time and memory cost (Section 5).

- ~~**Two-tier evaluation at real payload sizes**~~**Experiments and validation.** ~~A real trainer covers non-IID CIFAR-10, CIFAR-100, and MNIST over ten seeds,~~ On a real federated LoRA trainer and a 500-client scheduler simulation~~covers deadline, budget, mobility, and scale regimes. Under a 0.1 s deadline~~, FedRAS delivers 22% more timely updates ~~and~~ with 23% fewer bytes than fixed rank 16 ~~at~~ under deadline stress at statistically equivalent accuracy, while ~~full-parameter~~ full-model updates miss every stress deadline~~at 58.7× the bytes. The evaluation further covers external baselines, a candidate-pool~~

~~decomposition of the scheduler's gains, a SAGIN link profile, and pre-registered equivalence tests.~~ ; the conclusions are robust across datasets, link profiles, and delivery-predictor distortions (Section 6). Accuracy, timely delivery, and bytes are reported as separate outcomes throughout.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents the system model, the adversary and trust assumptions, and the scheduling formulation. Section 4 describes the FedRAS design and Section 5 its theoretical analysis. Section 6 reports the experimental evaluation, Section 7 discusses interpretation, deployment, and limitations, and Section 8 concludes.

A preliminary poster version of this work studied joint client–protection scheduling for federated learning in space–air–ground mobile networks [22]. The present article substantially reformulates that line of work around a different schedulable resource. It replaces compression-and-protection policies with the LoRA rank as the per-client decision variable. It introduces the real-payload delivery hazard and the saturating capacity model of Section 3, replaces the heuristic admission-and-repair procedure with a dynamic program that is exact on the stated budget grid, and adds frozen-factor rank-masked aggregation for heterogeneous ranks. It also reports a new two-tier evaluation comprising a real federated LoRA trainer, a 500-client scheduling simulator, and a solver exactness study. Accordingly, all numerical results in this article come from the revised formulation and implementation and are not comparable with the preliminary version.

~~We report accuracy, timely delivery ratio, and bytes per round separately. With a frozen pretrained backbone, the expected benefit of rank selection lies on the network side rather than in universal accuracy improvement.~~

~~The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 introduces the system model. Section 3.1 formulates the scheduling problem and Section 4 presents the solver and the aggregation scheme. Section 6 reports the evaluation, Section 7 discusses limitations, and Section 8 concludes.~~

2. Related work

2.1. ~~*Parameter-efficient and federated*~~ *Federated LoRA and heterogeneous-rank adaptation*

LoRA freezes a pretrained backbone and learns low-rank increments $\Delta W = BA$ [11]. Federated LoRA trains these adapters across clients. Prior work has studied sparse updates ~~, instruction tuning , and model-heterogeneous personalization [12, 13, 20]~~and instruction tuning [12, 13], examined how standard parameter-efficient tuning behaves under the federated setting [15] , and pursued personalization through model-heterogeneous or dual-adapter designs [20, 21]. A further line lets the rank vary across clients and aligns heterogeneous ranks for aggregation, by rank self-pruning with sparsity-weighted averaging [16], stacking [23], zero-padded rank alignment [17], or SVD-based dynamic rank allocation [18].

FFA-LoRA [14] froze one factor at its random initialization and showed that this improves stability in privacy-preserving federated fine-tuning. We adopt the same frozen-factor property for a stronger reason. When the frozen factor is identical across clients, averaging the trainable factor gives exact coordinate-wise averaging of the active induced-update slices, as Proposition 3 shows, and each client uploads only that factor. It thereby removes the factor-rotation ambiguity that makes joint factor averaging differ from update averaging. Section 4.2 develops this aggregation.

SA-FedLoRA schedules rank over rounds without conditioning on link state [19]. FedSA-LoRA studies asymmetric factor aggregation with client-local parameters [24]. ~~These rank-adaptation rules respond~~ Across these systems, the rank responds to training progress, parameter importance, or device capability~~, not to~~ ; in none of them is it a decision variable conditioned on the delivery state of a deadline-constrained link. We evaluate an annealing baseline and joint A/B averaging while making rank a per-round decision conditioned on delivery state and deadline.

2.2. *Communication-efficient federated learning*

A broad literature reduces FL communication through sparsification ~~, quantization, compressed sensing, and~~ [25], quantization [26, 27], quantized compressed sensing [28, 29], and beamformed over-the-air aggregation ~~[25–30]~~[30]. These methods compress a given update. Our scheduler selects its rank before transmission, so rank directly affects payload and

deadline feasibility. The proposed capacity and delivery models can therefore be combined with other codecs; Section 6.2.1 validates this composition with an FP16 upload.

2.3. Client selection and deadline-aware scheduling

Client selection improves FL efficiency under data, latency, and energy heterogeneity~~[31–34]. Deadline-~~. Oort ranks candidates by a utility that couples data value with system efficiency [31]; FedCS selects clients under explicit resource constraints [35]; TiFL groups clients into latency tiers to bound stragglers [36]; further designs target cost and freshness [32–34]. A second line makes network scheduling deadline- and reliability-aware ~~participation has also been studied~~ in mobile and non-terrestrial networks~~[7–10, 37]~~, from client participation [7, 8, 10, 37] to traffic offloading driven by federated reinforcement learning [9]; LESSON admits wireless clients at differentiated frequencies under a latency budget [38], and FedSN coordinates federated training across heterogeneous LEO satellite networks [39]. These methods generally treat update size as fixed: participation is decided for a given payload, not jointly with it. We include an Oort-style selector as a selection baseline. FedRAS chooses participation and update rank together, yielding a multiple-choice knapsack.

2.4. ~~Positioning~~Research gap and positioning

Across these lines, a common interface is missing. The rank-adaptation methods of Section 2.1 decide the rank from training state, parameter importance, or device capability; the selection and deadline-aware methods of Section 2.3 decide participation for a fixed update size. Neither treats the LoRA rank as a schedulable network resource whose payload sets the delivery probability on a lossy link, and none formulates the joint client–rank assignment under byte-budget and deadline constraints. Table 1 positions FedRAS against the closest systems along these dimensions.

We do not claim each ingredient individually as new. Our contribution is their combination~~. It comprises~~ : the LoRA rank treated as a deadline-constrained network resource with an explicit payload-conditioned delivery hazard, a client–rank assignment that is exact on the stated budget grid ~~,~~ with quantified greedy gaps and solver cost, and validation ~~of the resulting trade-offs~~ with a real federated trainer in which adapters are placed on every block of the backbone and all byte accounting uses real serialized sizes. The theoretical contribution is mechanism correctness rather

Table 1: Positioning against federated LoRA and client-selection systems. ✓ = the property holds in the cited papers' own descriptions; × = absent or not reported. Delivery-aware means that a scheduling or adaptation decision conditions on per-round network delivery state (rate, deadline, delivery probability) rather than on training progress, device capability, or static profiling alone. Real payload means payload is accounted as transmitted bytes rather than a fixed or parameter-count proxy. Real training means model training is executed in the evaluation as reported.

System	Var. rank	Deliv.-aware	Het.-rank agg.	Admis-sion	Joint adm.+rank	Real payload	Real training
FFA-LoRA [14]	×	×	×	×	×	×	✓
SLoRA [12]	×	×	×	×	×	×	✓
FedDPA [21]	×	×	×	×	×	×	✓
HetLoRA [16]	✓	×	✓	×	×	×	✓
HeLoRA [17]	✓	×	✓	×	×	×	✓
FedARA [18]	✓	×	✓	×	×	×	✓
SA-FedLoRA [19]	✓	×	×	×	×	×	×
FedSA-LoRA [24]	×	×	×	×	×	×	✓
Oort [31]	×	×	×	✓	×	×	✓
FedCS [35]	×	✓	×	✓	×	×	✓
TiFL [36]	×	×	×	✓	×	×	✓
LESSON [38]	×	✓	×	✓	×	×	✓
FedSN [39]	×	✓	×	✓	×	×	✓
FedRAS (ours)	✓	✓	✓	✓	✓	✓	✓

than a learning-performance guarantee: exactness of the assignment on the stated grid (Proposition 1) and exactness of frozen-factor aggregation (Proposition 3), each with its scope conditions stated where it applies. The claim is conditional on the surveyed literature of Section 2 and on the evaluated setting.

3. System model and threat model

3.1. ~~Mobile edge FL~~ Mobile-edge federated learning architecture and round protocol

We consider a mobile edge network in which an edge aggregator serves a client set $\mathcal{D} = \mathcal{D}_g \cup \mathcal{D}_a \cup \mathcal{D}_s$ of ground clients $\mathcal{D}_g$, aerial and UAV clients $\mathcal{D}_a$, and satellite-connected clients $\mathcal{D}_s$. Ground links are comparatively stable, aerial links face frequent handover, and satellite links suffer long propagation delay, intermittent visibility, and outages [4, 7]. The aggregator

holds the global model, schedules every round, and aggregates the timely updates. ~~Fig. 1 shows the resulting round pipeline. In every round the aggregator estimates the link states, constructs and filters the client–rank actions, and solves the assignment with the grid-based dynamic program of Section 4. It then executes the federated round and feeds the observed delivery outcomes back into the link-state history. The experiments evaluate both a homogeneous link profile and a grouped ground–aerial–satellite profile with distinct rate and mobility laws (Section 6).~~

3.2. ~~Federated LoRA rounds~~

Let w^t denote the frozen backbone broadcast at round t. Client i trains LoRA factors $A_{i,\ell}$ and $B_{i,\ell}$ for the adapted layers $\ell \in \mathcal{L}$~~. The~~ , with effective weight change ~~is~~ $\Delta W_{i,\ell} = B_{i,\ell}A_{i,\ell}$. A rank r update uses the first r columns of $B_{i,\ell}$ and the matching rows of the shared A. ~~A round then proceeds as follows.The aggregator broadcasts the backbone and the current adapter state. It observes the client link states~~Fig. 1 shows the resulting round pipeline. In every round, the aggregator estimates the link states, scores the client–rank actions with Action Construction (AC, Section 4.2), and solves the ~~client–rank assignment under the deadline, budget, and participation constraints. The~~ assignment with the GridDP assignment (GDP, Section 4.1); the admitted clients train locally with their factors ~~truncated~~ sliced to the assigned rank, so only the first r coordinates receive gradient updates, and upload before the deadline τ~~. The aggregator aggregates~~ ; the aggregator then merges the timely updates with ~~data-size weights~~ Masked Aggregation (MA, Section 4.2) and updates its link-state history from the observed outcomes. Late or lost updates are discarded, because stale updates harm convergence stability in deadline-critical federated learning. The experiments evaluate both a homogeneous link profile and a grouped ground–aerial–satellite profile with distinct rate and mobility laws (Section 6).

3.2. Link, mobility, and ~~delivery~~ timely-delivery model

At decision time the scheduler receives an estimated uplink rate $\bar{R}_i^t$ in Mbps and a mobility state. The realized rate R_i^t, handover event $H_i^t \in \{0, 1\}$, and packet loss indicator J_i^t are not observed before transmission. For an update of $B_i^t(r)$ megabytes, the modeled delivery time is

$$T_i^t(r) = T_{\text{train,i}}(r) + T_{\text{proc,i}}(r) + \frac{8B_i^t(r)}{R_i^t} + d_i^t + H_i^t \Delta d_h. \tag{1}$$

Timely delivery is the event

$$P^t_{\mathrm{on,i}}(r) = \Pr\big(T^t_i(r) \le \tau,\ J^t_i = 0 \mid x^t_i\big)\,. \tag{2}$$

The implementation uses the following hazard surrogate rather than a derived distribution for all terms in (1).

$$\hat{P}^t_{\mathrm{on,i}}(r) = \exp\big(-\mu^t_i\big)\exp\Big(-\frac{8B^t_i(r)}{\bar{R}^t_i\tau}\Big). \tag{3}$$

The dimensionless intensity μ^t_i summarizes mobility-related failure risk, packet loss, handover effects, and unmodeled processing time~~. It is not estimated by an EWMA~~; it is a fixed surrogate rather than an estimated distribution in the present implementation~~. The experiments generate mobility state from the stated Beta model and provide that state and the generated rate as oracle estimates. Thus, (3) is an idealized~~, and (3) is used as a plug-in predictor ~~used to isolate scheduling behavior. The calibration and noise studies assess sensitivity to this predictor, but they do not constitute calibration on measured wireless traces. Training and processing times are included in the conceptual delivery model but are not separately measured in the current experiments~~throughout. The construction of the estimated states and the sensitivity studies are specified in Section 6; how far the conclusions survive this idealization is delimited in Section 7.

3.3. Rank-dependent ~~update size~~ payload and adaptation capacity

With b bits per parameter and the frozen factor A shared and never transmitted, a rank-r client uploads only the trainable factor B, whose per-rank cost is the summed output dimension $D = \sum_{\ell\in\mathcal{L}} d_\ell$ of the adapted layers. The real serialized payload is

$$B^t_i(r) = \frac{b}{8}\, r\, D + B_{\mathrm{meta}}, \tag{4}$$

in bytes~~, with $D = 1{,}408$ parameters per rank column and $b = 32$ in the evaluated CNN, giving $B^t(16) = 0.090$ MB when the header allowance B_{meta} is set to zero in the metering (Table 2)~~. Rank is therefore a network-visible resource~~. It~~: it scales both the adaptation capacity and the transmission time, and its per-unit payload is audited against the evaluated model in Section 6. We model the useful contribution of a delivered rank-r update with the saturating capacity gain

$$G^t_i(r) = \pi^t_i\,\sqrt{\min(r, r_{\max})/r_{\max}}, \tag{5}$$

where π_i^t is a per-client contribution prior in the spirit of Oort-style utility [31]. The width $r_{\max}$ is the factor width provisioned at all clients. The square root encodes the diminishing returns of rank ~~; the trainer data support this shape, with the mean delivered update norm growing by about 1.9× from rank 4 to rank 16 against the 4× of a linear model. The~~ that the delivered-update norms exhibit directly in the trainer (Section 6.2), and the saturation at $r_{\max}$ reflects that wider factors add payload but no expressible capacity once the provisioned width is reached. Full-parameter FedAvg updates are accounted at their true serialized size ~~, 5.29 MB in the evaluated model, i. e. , 58.7× a rank-16 upload~~(Table 2).

3.4. Adversary and trust model

Clients are honest but curious: they follow the training protocol, and raw data never leaves the devices, which is the standing premise of federated learning on this infrastructure. The edge aggregator is trusted scheduling infrastructure: it observes link states, client–rank assignments, and delivered update factors, and runs the aggregation of Section 4.2. The external adversary of interest is an eavesdropper on the wireless uplink who observes the transmitted B factors. Because the frozen factor A is public and a rank-r client transmits only its first r coordinates, the object reconstructable from an intercepted transmission is the rank-r induced update, and the exposed surface is its B-slice: a transmission strictly smaller in bytes and in exposed coordinate count than a joint-factor or full-model update. This is a reduction in exposed transmission, not a protection mechanism; FedRAS proposes none, and differential-privacy accounting, secure aggregation, and Byzantine robustness are orthogonal and out of scope (Section 7).

4. ~~Deadline-aware client–rank scheduling~~

3.1. Deadline-constrained client–rank scheduling formulation

Each round, the aggregator decides $a_i^t \in \{0\} \cup \mathcal{R}$, where $a_i^t = 0$ means client i does not participate and $a_i^t = r$ schedules it at rank r. The per-round problem is

$$\max_{\{a_i^t\}} \sum_{i \in \mathcal{D}} \mathbf{1}\{a_i^t > 0\}\, G_i^t(a_i^t)\, P_{\text{on,i}}^t(a_i^t) \tag{6}$$

$$\text{s.t.} \quad \sum_i B_i^t(a_i^t) \le B_{\max}, \qquad \sum_i \mathbf{1}\{a_i^t > 0\} \le K. \tag{7}$$

The objective is a scheduling utility~~. It counts~~ : expected useful delivered updates under the byte budget $B_{\max}$ and the participation cap K. It is ~~not~~ a ~~direct~~ network-side objective, not a surrogate for final test accuracy~~. Increasing~~ ; increasing r raises G sublinearly ~~, increases payload , and lowers~~ while raising payload and lowering the delivery surrogate. The experiments therefore report test accuracy, timely delivery ratio, bytes per round, and mean delivered rank as separate outcomes (Section 6).

~~Two deployment variants are evaluated. In full grid scheduling~~

4. FedRAS design

4.1. Overview

Algorithm 1 summarizes one scheduling round as a fixed sequence of three named components. Action Construction (AC, Section 4.2) scores every client–rank pair with a real payload, a delivery probability, and a capacity gain; the GridDP assignment (GDP, Section 4.1) solves the scored assignment exactly under the byte budget and the admission cap, with a cheap per-client rule covering the unconstrained regime; and Masked Aggregation (MA, Section 4.2) merges heterogeneous-rank updates toward the frozen shared factor. Fig. 1 places AC, GDP, and MA in the round pipeline. The design contains no training-side logic: admissions and ranks are chosen from network state alone.

4.2. Action Construction (AC)

Each round the aggregator selects, for every client, one action from the discrete set $a_i^t \in \{0\} \cup \mathcal{R}$ of Section 3.1. AC builds and scores this action set (Algorithm 2): the per-rank payload follows (4), the delivery hazard (3) and the capacity gain (5) give the estimated utility $u_i(r) = G_i^t(r)\hat{P}_{\mathrm{on},i}^t(r)$, and a conservative ceiling discretization onto a ~~0.01 MB grid , applied identically~~ budget grid q yields the integer cost $\widetilde{B}_i(r) = \lceil B_i^t(r)/q \rceil$. The same grid is applied to the dynamic program, the greedy baselines, and the admission checks, so that every scheduling method optimizes over the identical feasible region. ~~Section 6.2.1 compares the variants in the trainer.~~

5. ~~Grid-based solver and heterogeneous-rank aggregation~~

4.1. Dynamic program

Algorithm 1 FedRAS Round

Require: clients $\mathcal{D}$, rank set $\mathcal{R}$, cap K, budget $B_{\max}$, deadline τ, ~~the aggregator solves (6) and (7) over all clients. The solver study evaluates this variant at 500-client scale,~~ link-state history $\mathcal{H}^{t-1}$

Ensure: assignment a^t; new adapter state S^t; updated history $\mathcal{H}^t$

1: estimate per-client link states x_i^t from $\mathcal{H}^{t-1}$
2: score all pairs (i, r): utilities $u_i(r)$ and integer costs $\widetilde{B}_i(r)$ (AC, Algorithm 2)
3: solve the assignment a^t under K and ~~a same-cohort variant restricts the dynamic program to a uniformly sampled cohort, isolating the value of candidate-pool widening from in-cohort rank allocation. In the real trainer, fair-cohort scheduling samples the fixed cohort of ten clients and applies the~~ $B_{\max}$ (GDP, Algorithm 3); when the all-$r_{\max}$ profile is budget-feasible, the per-client ~~rank rule $r_i^\star = \arg\max_r G_i^t(r)\hat{P}_{\text{on,i}}^t(r)$. All budget arithmetic uses~~ maximizer rule is already optimal (Proposition 2)
4: broadcast the backbone w^t; admitted clients train locally with factors sliced to a_i^t and upload before τ
5: pass uploads arriving within τ (on-time gate)
6: aggregate timely updates with the frozen shared A (MA, Algorithm 4)
7: append observed outcomes (timely, late, lost) to $\mathcal{H}^t$

~~Problem~~ Under this discretization, the single-round estimated problem (6)–(7) is a multiple-choice knapsack with an admission constraint~~. Let $q = 0.01$ MB be the budget grid and define the integer cost $\widetilde{B}_i(r) = \lceil B_i^t(r)/q \rceil$~~, for which an exact pseudo-polynomial dynamic program exists; the evaluated grid width and its quantization error are quantified in Section 6.3.1.

4.1. GridDP Assignment (GDP)

GDP solves the scored assignment exactly (Algorithm 3). Let $V_i(k, b)$ be the maximum estimated utility from the first i clients with k admissions and integer budget b. Initialize $V_0(0, 0) = 0$ and all other states to $-\infty$. The recurrence is

$$V_i(k, b) = \max\left\{V_{i-1}(k, b),\ \max_{r\in\mathcal{R}}\left[V_{i-1}(k-1, b-\widetilde{B}_i(r)) + u_i(r)\right]\right\}, \qquad (8)$$

where $u_i(r) = G_i^t(r)\hat{P}_{\text{on,i}}^t(r)$. Backpointers recover the assignment. Propositions 1 and 2 (Section 5) establish the exactness of this recurrence and the regime

Algorithm 2 Action Construction (AC)

Require: clients $\mathcal{D}$, ranks $\mathcal{R}$, budget grid q, link states $\{x_i^t\}$
Ensure: scored action set $\{u_i(r), \widetilde{B}_i(r)\}_{i\in\mathcal{D}, r\in\mathcal{R}}$

for each client i and rank r **do**
  payload $B_i^t(r)$ by (4); integer cost $\widetilde{B}_i(r) \leftarrow \lceil B_i^t(r)/q \rceil$
  delivery probability $\hat{P}_{\text{on},i}^t(r)$ by (3); capacity $G_i^t(r)$ by (5)
  utility $u_i(r) \leftarrow G_i^t(r)\, \hat{P}_{\text{on},i}^t(r)$
end for
return the scored set; pairs with $\widetilde{B}_i(r) > \widetilde{B}_{\max}$ are infeasible

Algorithm 3 GridDP Assignment (GDP)

Require: scored actions $\{u_i(r), \widetilde{B}_i(r)\}$ from AC, cap K, budget grid $\widetilde{B}_{\max}$
Ensure: client–rank assignment a^t

$V_0(0,0) \leftarrow 0$; all other DP states $\leftarrow -\infty$
for each client $i \in \mathcal{D}$ **do**
  for $k = 0, \ldots, K$ and each budget state b **do**
    $V_i(k,b) \leftarrow$ skip or best rank transition of (8)
    store the maximizing action as a backpointer
  end for
end for
backtrack from $\max_{k,b} V_{|\mathcal{D}|}(k,b)$; return a^t

in which the per-client maximizer rule $r_i^\star$ is already optimal, in which case GDP reduces to that rule at zero cost.

4.2. Masked Aggregation (MA)

Clients scheduled at different ranks share an $r_{\max}$ coordinate system. For one adapted layer, client i has sample weight n_i and delivered rank r_i. The factor $A \in \mathbb{R}^{r_{\max}\times k}$ is frozen and shared, while $B_i \in \mathbb{R}^{d\times r_{\max}}$ is trainable. A rank r_i update activates columns 1 through r_i of B_i and the matching rows of A, giving $\Delta W_i = \sum_{j\le r_i} [B_i]_{:,j}[A]_{j,:}$.

Jointly trained factor pairs (A_i, B_i) are identifiable only up to invertible transforms, so separately averaged factors need not average the updates at all. The elevated-learning-rate ablation of Section 6 examines this aggregation rule empirically.

Algorithm 4 Masked Aggregation (MA)

Require: timely updates $\{(n_i, B_i, r_i)\}$, frozen shared A, momentum β, previous state S^{t-1}

Ensure: new adapter state S^t

1: **for** each rank column j of each adapted layer **do**
2: $\quad W_j \leftarrow \sum_{i:r_i \geq j} n_i$; $\quad \bar{B}_{:,j} \leftarrow \frac{1}{W_j} \sum_{i:r_i \geq j} n_i [B_i]_{:,j}$
3: **end for**
4: $\bar{A} \leftarrow A$
5: $\Delta^t \leftarrow \bar{B} - S^{t-1}$; $\quad v^t \leftarrow \beta\, v^{t-1} + (1-\beta)\, \Delta^t$; $\quad S^t \leftarrow S^{t-1} + v^t$

Rank-masked averaging. The server zero-pads inactive columns of each delivered B_i and averages each active coordinate over the clients that activate it,

$$\bar{B}_{:,j} = \frac{1}{W_j} \sum_{i:\, r_i \geq j} n_i \left[B_i\right]_{:,j}, \qquad W_j = \sum_{i:\, r_i \geq j} n_i, \tag{9}$$

while the frozen factor aggregates to itself, $\bar{A} = A$. The server then applies momentum per factor tensor. With $\Delta^t = \bar{B} - S^{t-1}$, the velocity and state update as $v^t = \beta\, v^{t-1} + (1-\beta)\, \Delta^t$ and $S^t = S^{t-1} + v^t$, with damping coefficient β absorbing client-sampling and link noise. Algorithm 4 fixes the procedure; Proposition 3 (Section 5) states its exactness and dilution properties.

5. Theoretical analysis

5.1. Exactness of the DP assignment

Proposition 1 (Per-round optimality)**.** *For every reachable state (k, b), recurrence* (8) *considers the skip action and every feasible rank for client i. Induction on i therefore proves exactness for the discretized single round estimated assignment problem. ~~The result is not a claim of optimality for the continuous budget problem, the multi-round stochastic control problem, or final model accuracy.~~*

Proof. The admission indicator and the rank of each client are exclusive actions that affect the two resources additively. Admitting client i at rank r consumes one admission slot and $\widetilde{B}_i(r)$ budget units and adds $u_i(r)$ value,

independent of the other clients. The state (k, b) therefore summarizes all interaction, and the standard interchange argument for the multiple-choice knapsack applies. Any optimal solution restricted to the first $i-1$ clients must itself be optimal for its state, which the recurrence maximizes. □

~~The complexity is $O(|\mathcal{D}|K\widetilde{B}_{\max}|\mathcal{R}|)$ time and $O(K\widetilde{B}_{\max})$ memory with rolling value arrays; the back-pointer table retained for reconstruction adds $O(|\mathcal{D}|K\widetilde{B}_{\max})$, where $\widetilde{B}_{\max} = \lceil B_{\max}/q \rceil$. Section 6.3.1 reports the resulting runtimes for 500 and 1000 clients. The exact~~ Two readings make the statement precise. First, exactness is relative to the discretized problem: the dynamic program returns the best schedule among those the budget grid can express, so any residual gap to the continuous optimum is a quantization error that the grid step controls and Section 6.3.1 measures. Second, and more importantly for practice, exactness localizes the sources of loss. Given the estimated delivery probabilities, no feasible schedule can score higher than the DP output; whatever gap remains between planned and realized performance therefore originates in prediction error or in the round-level abstraction, not in the solver. This separation is what makes the calibration study of Section 6.3.2 interpretable: it perturbs the predictor and observes a smooth, bounded objective loss precisely because the optimization layer contributes no loss of its own.

5.2. Optimality condition of the per-client rule

The formulation also yields a condition under which the cheap per-client rule ~~of Section 3.1~~ — admit every candidate and assign each its own maximizer $r_i^\star = \arg\max_{r\in\mathcal{R}} G_i(r)\hat{P}_{\mathrm{on},i}(r)$ — is already optimal, which the trainer experiments of Section 6.2.1 rely on.

Proposition 2 (Cohort-local optimality of the per-client rule)**.** *Consider the single-round problem* (6)–(7) *restricted to a fixed candidate set of at most K clients, and suppose the all-$r_{\max}$ rank profile is budget-feasible, i.e., $\sum_i \widetilde{B}_i(r_{\max}) \le \widetilde{B}_{\max}$ over the candidates. Then the profile that assigns every admitted client its per-client maximizer $r_i^\star = \arg\max_{r\in\mathcal{R}} G_i(r)\hat{P}_{\mathrm{on},i}(r)$ is a feasible optimal solution of the restricted problem.*

Proof. Each candidate's cost is at most its $r_{\max}$ cost, so the per-client maximizer profile satisfies the budget whenever the all-$r_{\max}$ profile does, and, with at most K candidates, it uses no more than the cap. Being feasible, it attains $\sum_i \max_r u_i(r)$, which upper-bounds the objective of every feasible

profile; it is therefore optimal. By Proposition 1, the dynamic program returns an optimal profile and hence coincides with the per-client maximizer profile whenever the latter is unique. □

FedRAS scheduling round clients $\mathcal{D}$, ranks $\mathcal{R}$, cap K, budget $B_{\max}$, deadline τ, states $\{x_i^t\}$ admitted set and assigned ranks Estimate $\hat{P}^t_{\mathrm{on},i}(r)$ by (3) and $G_i^t(r)$ by (5) for all i, r Initialize $V_0(0,0) = 0$; other DP states to $-\infty$ Apply skip and every feasible ranktransition of (8) Store the maximizing action as a backpointer Backtrack from $\max_{k,b} V_{|\mathcal{D}|}(k,b)$; return the client–rank assignment Execute the round; aggregate timely rank-masked updates (Section 4.2); update link-state history

5.3. Rank-masked aggregation with frozen factors

Clients scheduled at different ranks share an $r_{\max}$ coordinate system. For one adapted layer, client i has sample weight n_i and delivered rank r_i. The factor $A \in \mathbb{R}^{r_{\max} \times k}$ is frozen and shared, while $B_i \in \mathbb{R}^{d \times r_{\max}}$ is trainable. A rank r_i update activates columns 1 through r_i of B_i and the matching rows of A, giving $\Delta W_i = \sum_{j \le r_i} [B_i]_{:,j} [A]_{j,:}$The condition has a direct economic reading: Proposition 2 holds when the budget is not scarce enough to force a sacrifice, that is, when every candidate can take its unconstrained favorite rank.

Jointly trained factor pairs (A_i, B_i) are identifiable only up to invertible transforms, so separately averaged factors need not average the updates at all. The elevated-learning-rate ablation of Section 6 examines this aggregation rule empirically.

Rank-masked averaging. The server zero-pads inactive columns of each delivered B_i and averages each active coordinate over the clients that activate it,

$$\bar{B}_{:,j} = \frac{1}{W_j} \sum_{i:\, r_i \ge j} n_i \left[B_i\right]_{:,j}, \qquad W_j = \sum_{i:\, r_i \ge j} n_i,$$

while the frozen factor aggregates to itself, $\bar{A} = A$. The server then applies momentum per factor tensor. With $\Delta^t = \bar{B} - S^{t-1}$, In that regime the objective decomposes across clients, local selfishness is globally optimal, and the velocity and state update as $v^t = \beta\, v^{t-1} + (1-\beta)\, \Delta^t$ and $S^t = S^{t-1} + v^t$, with $\beta = 0.3$ damping client-sampling andlink noisescheduler reduces to $|\mathcal{D}|$ table lookups. Greedy rules fail exactly when this condition breaks: once the budget forces trade-offs between clients, the action that is best for one

client changes the residual budget available to the others, and no per-client ordering resolves that coupling. The condition is also actionable at run time. Checking $\sum_i \widetilde{B}_i(r_{\max}) \le \widetilde{B}_{\max}$ costs one pass over the candidates, so whenever the candidate pool already fits within the cap K the scheduler can run the cheap rule by default and invoke the dynamic program only when the budget check fails or the pool must be culled below the cap, which is precisely where admission coupling appears — the deployment rule exercised in Sections 6.2.1 and 7.

5.3. Exactness of frozen-factor aggregation

Proposition 3 (Exactness of frozen-factor aggregation)**.** *Under* (9)*, the j-th active rank slice of $\bar{B}A$ is the data-weighted average over clients with $r_i \ge j$. This is exact coordinate-wise averaging of active induced-update slices. It is not, in general, the unified data-weighted average of complete client updates. If a common denominator $W = \sum_i n_i$ is used instead of W_j, the unified average is recovered, but active high-rank slices are diluted by W_j/W. The two rules coincide for all possible updates when all delivered ranks are equal.*

Proof. The first two parts follow from linearity. For frozen shared A, multiplying a weighted sum of B-coordinates by $[A]_{j,:}$ converts it into the same weighted sum of the matching update slices, and the two normalizers differ only in W_j versus W. The product $\bar{B}A = \sum_j \bar{B}_{:,j}[A]_{j,:}$ collects the slices. The third part holds because $W_j = W$ for every active j if and only if $r_i = \max_k r_k$ for all admitted i with $n_i > 0$. □

The ~~proposition concerns active rank slices rather than whole-update equivalence. With shared frozen~~ frozen factor is what makes heterogeneous ranks commensurable. If both factors rotate per client, the product structure of an update is client-specific and no server-side average of factors corresponds to an average of updates; with A ~~, it gives coordinate-wise~~ shared, rank slicing acts on B alone and the j-th slice of every delivered update lives in the same coordinates. The two normalizers then embody a genuine choice. The denominator W_j preserves per-slice fidelity: each client's influence on the coordinates it actually sent is proportional to its data. A common denominator W recovers the FedAvg-style data-weighted ~~averaging and avoids dilution caused by a common denominator. The elevated-learning-rate ablation tests the associated stability claim. It does~~ average but dilutes the slices that only high-rank clients contribute, by the

ratio W_j/W. Which is preferable depends on whether one reads the round as averaging updates or as averaging clients; FedRAS takes the former reading, and the two rules coincide whenever delivered ranks do not mix, which the nominal-deadline regime of the experiments satisfies in practice.

5.4. *Complexity and scope of the analysis*

The complexity is $O(|\mathcal{D}|K\widetilde{B}_{\max}|\mathcal{R}|)$ time and $O(K\widetilde{B}_{\max})$ memory with rolling value arrays; the back-pointer table retained for reconstruction adds $O(|\mathcal{D}|K\widetilde{B}_{\max})$, where $\widetilde{B}_{\max} = \lceil B_{\max}/q \rceil$. Section 6.3.1 reports the resulting runtimes at 500- and 1000-client scale.

Scope of the analysis. The exactness results concern the discretized single-round estimated assignment problem and the frozen-factor aggregation rule. They are not claims of optimality for the continuous-budget problem, for the multi-round stochastic control problem, or for final model accuracy, and they do not establish a ~~general convergence guarantee~~ convergence guarantee for federated training. The practical value of exactness is quantified experimentally in Sections 6.2.1 and 6.3.1.

6. ~~Performance~~ Experimental evaluation

The evaluation tests accuracy and network behavior in a real trainer (Section 6.2), scheduling quality at 500 clients ~~, solver cost~~ and solver cost (Section 6.3.1), and sensitivity to payload, predictor, capacity, and budget-grid assumptions (Sections 6.3.1 and 6.3.2). All byte figures are real serialized sizes, and all deadline regimes are calibrated to those sizes.

6.1. *Experimental setup*

~~Real trainer~~Environment. The trainer is implemented in PyTorch~~and runs on one~~ , and all real-training runs use one NVIDIA A800 GPU.

Datasets and partitions. Experiments use CIFAR-10, CIFAR-100, and MNIST. The real trainer partitions 20 clients non-IID with Dirichlet $\alpha = 0.5$ and samples 10 participants per round; variations of α and of the mobility distribution are reported in Table 3(b) and Fig. 4.

Backbone and adapters. The backbone is a six-convolution BatchNorm CNN with channel progression 64/128/256, followed by two 256-wide linear layers. It is centrally pretrained with crop, flip, and erase augmentation, Nesterov momentum, weight decay, and cosine learning-rate decay, for 30 epochs on CIFAR-10, 45 on CIFAR-100, and 3 on MNIST, and then

frozen, reaching a test accuracy of 0.914, 0.693, and 0.995, respectively. LoRA adapters with candidate rank set $\mathcal{R} = \{2, 4, 8, 16\}$ and provisioned width $r_{\max} = 16$ are placed on all six convolutions through 1×1 factors and on both head layers. A ; a client assigned rank r trains with its factors sliced to the first r coordinates, so the rank genuinely bounds local adaptation capacity. The pretrained backbone reaches a test accuracy of 0.914 on CIFAR-10, 0.693 on CIFAR-100, and 0.995 on MNIST.

Federated protocol and linkstraining hyperparameters. The real trainer uses 20 clients partitioned with Dirichlet $\alpha = 0.5$, 10 sampled participants per round, Federated training runs for 40 rounds with one local epoch , batch size 128, and 40 rounds. and batch size 128. Local SGD uses learning rate 0.005, weight decay 10^{-4}, gradient clipping at 1, and cosine decay. The server uses applies rank-masked weighted averaging with momentum 0.3. Results use five seeds (0 through 4); the CIFAR-10 nominal and 0.1 s regimes, their external baselines, and the scheduling-arm comparison at the nominal deadline use ten seeds. β=0.3, the coefficient of Section 4.2.

Link models. Uplink rates are drawn from LogNormal$(2.5, 0.7)$ clipped to $[1, 40]$ Mbps and mobility from Beta$(2, 5)$, redrawn each round; a . The mobility draw is the hazard intensity μ_i^t of (3) itself: the implementation simulates no separate propagation, processing, handover-delay, or packet-loss process, and their contribution is absorbed into μ_i^t; the realized delivery of each update is a Bernoulli draw with probability (3) at the drawn state, so the deployed predictor is also the ground-truth generator and the estimates are oracle (Section 7 delimits this). The grouped ground–aerial–satellite profile with distinct rate and mobility lawsper class is evaluated separately. These generated states are supplied as oracle estimates to the schedulerassigns 60/25/15% of clients to classes with distinct laws: ground LogNormal$(2.8, 0.5)$ clipped to $[5, 40]$ Mbps with Beta$(2, 8)$ mobility, aerial LogNormal$(2.3, 0.7)$ clipped to $[2, 40]$ Mbps with Beta$(2, 5)$, and satellite LogNormal$(1.6, 0.5)$ clipped to $[0.5, 15]$ Mbps with Beta$(2, 3)$.

Deadline calibration. All deadline regimes are calibrated to the real payloads. The nominal round deadline is 2.0 s, at which transmission time consumes only 3% of a rank-16 update's delivery probability at the median 12 Mbps rate, while a full-model update already misses 83% of its deadlines. The stress deadlines are 0.1 s and 0.05 s, at which rank-16 timely delivery falls to 0.41 and 0.27 while rank-4 retains 0.62 and 0.57, so downshifting is genuinely load-bearing. A loose 10 s control restores full-model timely

Table 2: Payload audit of real serialized update sizes. Frozen-A clients upload only the trainable B factor (1,408 parameters per rank column); the A+B column is the joint-training equivalent. The stylized $s(r)$ column lists the ~~normalized-cost~~ protocol reused as a normalized-cost cell in ~~the sensitivity study of~~ Section 6.3.1.

Rank r	A+B FP32 (MB)	B-only FP32 (MB)	B-only FP16 (MB)	Stylized $s(r)$ (MB)
2	0.021	0.011	0.006	0.54
4	0.041	0.023	0.011	0.58
8	0.082	0.045	0.023	0.66
16	0.164	0.090	0.045	0.82
Full model	5.29 MB FP32 (58.7× the rank-16 B-only upload)			

delivery to 0.52.

Model and payload audit. ~~Table 2 audits the actual serialized sizes against the evaluated model.~~ The frozen backbone has 1.28M parameters~~(5.12 MB in FP32). Each rank column carries 2,563 adapter parameters across the eight adapted layers, of which the frozen A contributes 1,155 and the transmitted~~ ; each adapted layer contributes a frozen-A and a trainable-B ~~contributes 1,408, i.e., 5.63 KB~~ slice per rank column~~. A rank-16 client therefore uploads 0.090 MB (FP32), and a full-model update transmits 5.29 MB, 58.7× more~~, and only B is transmitted. Table 2 lists the audited serialized sizes, and every deadline regime of this study is calibrated against them. The metadata term B_{meta} of (4) is zero in both evaluation tiers, since the server broadcast is not part of the uplink accounting.

Simulator. The 500-client, 150-round scheduler simulation uses the same delivery and capacity models with uniform cohort sampling, per-client rank selection, and budget admission on the same ceiling grid; its payload table (5,632 B per rank column) is identical to the trainer's, and its MB column reports grid-billed payloads on that same ceiling (a rank-16 update is billed 0.10 MB against a true 0.090 MB), since admission and accounting share the discretization. It covers nominal, deadline, budget, mobility, and 1000-client scalability regimes, with five seeds and common traces. Its accuracy column is a bounded scheduling proxy driven by delivered useful value, not a dataset accuracy; the full-update baseline receives the same capacity credit as rank 16 under the saturating model, a disclosed modeling assumption. A separate cell repeats the earlier stylized protocol ($s(r) = 0.02r + 0.5$ MB, rank-64 full-update stand-in, 0.25 s deadline, 12 MB budget) as a normalized-cost sensitivity check.

Scoring proxy. The simulator samples π_i uniformly from $[0.15, 0.45]$ and

uses a bounded scheduling proxy rather than dataset accuracy. Starting at 0.1, it updates as $a^{t+1} = a^t + \eta\hat{g}^t(1-a^t) - \varepsilon a^t + \omega^t$, where $\hat{g}^t$ is useful delivered value normalized by K, $\eta = 0.08$, $\varepsilon = 0.0005$, and $\omega^t \sim \mathcal{N}(0, 5 \times 10^{-4})$ is a small tie-breaking noise. No per-client prior is measured in the trainer experiments, so the trainer capacity term sets π_i=1; the sampled π_i enters only the simulator objective.

Solver study. The solver study runs 30 instances per deadline and budget cell at 500 clients, on a q=0.01 MB ceiling grid. It compares the grid-based DP with value-per-byte greedy, value greedy, a cohort-myopic rule, and fixed rank 16, and adds a client-count sweep that extends to 1000 clients. Budgets are 30 MB (slack), 4 MB, and 1.2 MB; the last admits only twelve rank-16 updates on the ceiling grid of the cap K=25 and is therefore binding.

~~All trainer methods use the same client partitions, local optimizer (the~~ **Baselines.** FedRAS is compared against fixed ranks r=4 and r=16, a random-rank policy, and full-model ~~baseline~~ FedAvg updates (referred to as full-model updates throughout, trained at a 0.4× local learning-rate scale~~), rank set where applicable, generated rate and mobility~~ ; at the shared 1.0× rate the head baseline loses about half a point of accuracy with transient minima near 0.89, so the reduced scale is favorable to the baseline and the comparison is conservative). Three external baselines are included: an SA-FedLoRA-style schedule that heats the rank from 2 to 16 and cools it down over the 40 rounds, oblivious to link state [19]; an Oort-style selector that samples $2K$ candidates and admits the top-K by a data-size-and-recency utility at fixed rank [31]; and joint A/B training with plain averaging, the aggregation-level competitor of Section 4.2 [24]. A delivery-only ablation (reliability-only) removes the capacity term. An offline control (FedAvg without delivery modeling) always aggregates its scheduled updates, bounding the capacity achievable without feasibility constraints. All methods share client partitions, generated link states, deadline, and seed protocol~~. The FedAvg baseline is referred to as full-model updates throughout. Communication metrics count attempted uplink bytes ,~~ ; where applicable they share the local optimizer and rank set.

Scheduling variants. Two deployment variants of the formulation are evaluated. In full grid scheduling, the aggregator solves (6) and (7) over all clients; the solver study evaluates this variant at 500-client scale, and a same-cohort variant restricts the dynamic program to a uniformly sampled cohort, isolating the value of candidate-pool widening from in-cohort rank

allocation. In the real trainer, fair-cohort scheduling samples the fixed cohort of ten clients and applies the per-client rank rule $r_i^\star = \arg\max_r G_i^t(r)\hat{P}_{\mathrm{on},i}^t(r)$. Section 6.2.1 compares the variants in the trainer.

Metrics. Four outcomes are reported throughout: test accuracy at round 40; timely delivery ratio, the timely deliveries of a method divided by its selected clients; bytes per round, counting attempted uplink payloads including late and failed uploads ~~. No quantization,~~ at real serialized sizes; and mean delivered rank over delivered updates. The simulator reports its bounded utility state, the mean delivery probability of admitted clients, and attempted payload totals (Table 4(a)). No sparsification, energy, or fairness metric is claimed in this study; a quantization baseline (FP16 upload) is evaluated in Section 6.2.1.

Statistical protocol. Results use five seeds (0 through 4) with common random numbers within each seed; the CIFAR-10 nominal and 0.1 s regimes, their external baselines, and the scheduling-arm comparison at the nominal deadline use ten seeds (0 through 9). Headline trainer comparisons average the final ten rounds (rounds 30–39). FedRAS versus fixed rank 16 is tested on ten paired seeds: the timely-delivery ratio improves in ten of ten pairs (two-sided Wilcoxon p=0.002), by +22% as the ratio of seed means with a bootstrap 95% confidence interval of the mean paired gain of [18%, 29%]; the attempted-byte saving is 23% (mean paired saving, [21%, 27%], Wilcoxon p=0.002). Accuracy equivalence uses a TOST test fixed in advance at a $\pm$0.5 percentage-point margin: the mean paired difference is +0.00 percentage points at the nominal deadline and +0.06 points under stress, with $p < 10^{-6}$ in both cases, so equivalence is established. Simulator comparisons without an accompanying test are descriptive.

6.2. ~~Accuracy parity and rank invariance in~~ Evaluation with the federated LoRA trainer~~Table 6~~

6.2.1. Accuracy parity and rank invariance

Table 3(a) reports ten-seed means and standard deviations after 40 federated rounds on CIFAR-10. All LoRA methods hold the backbone accuracy within one standard deviation of 0.914, whether the rank is fixed at 4, fixed at 16, drawn at random, or selected by the rank rule. Full-parameter updates spend ~~58.7× the bytes of a rank-16 update, and they~~ orders of magnitude more bytes per round (Table 2) and lose roughly three quarters of their timely deliveries ~~, 0.17 against 0.73, because their 5.29 MB~~ because their payloads rarely fit the round deadline. The cross-dataset results in Fig. ~~6~~

Table 3: ~~Real trainer on~~ Real-trainer experiments (non-IID CIFAR-10 ~~at the nominal 2.0 s deadline with ten seeds~~unless stated). Accuracy is ~~measured at round 40. Timely~~ round-40 test accuracy; timely delivery ratio~~is timely deliveries divided by selected clients. Bytes~~, bytes per round ~~is the sum of~~ (rounds 30–39, attempted uplink~~payloads for selected clients over rounds 30 to 39~~), ~~at real serialized sizes. Mean rank is computed over~~ and mean delivered ~~updates~~rank follow the definitions of Section 6. (a) Deadline ladder. (b) Extended stress matrix, five seeds. (c) External baselines, ten seeds except the two FP16 rows (five seeds). (d) Scheduling arms, 1.0 MB budget. (e) Component ablations, five seeds.

(a) Deadline ladder

	nominal 2.0 s				deadline 0.1 s				deadline 0.05 s			
Method	Acc.	Timely	MB	Rank	Acc.	Timely	MB	Rank	Acc.	Timely	MB	Rank
FedAvg (full model)	.9135 ± .0027	0.17	52.86	—	.9153 ± .0015	0.00	52.86	—	.9147 ± .0016	0.00	52.86	—
FedLoRA r=4	.9154 ± .0016	0.75	0.23	4.0	.9154 ± .0016	0.62	0.23	4	.9144 ± .0015	0.57	0.23	4
FedLoRA r=16	.9153 ± .0015	0.73	0.90	16.0	.9149 ± .0019	0.41	0.90	16	.9143 ± .0020	0.27	0.90	16
Random rank	.9151 ± .0016	0.76	0.42	7.3	.9153 ± .0017	0.56	0.42	6.2	.9143 ± .0015	0.46	0.42	5.7
FedRAS (ours)	.9153 ± .0016	0.73	0.90	16.0	.9154 ± .0018	0.49	0.69	12.7	.9143 ± .0014	0.49	0.44	8.1

(b) Extended stress matrix (round-40 accuracy)

Scenario	FedAvg	r=4	r=16	Random	FedRAS
deadline 10 s	0.9134	0.9144	0.9145	0.9141	0.9143
mobility 0.7	0.9115	0.9143	0.9141	0.9145	0.9141
α=0.1	0.9022	0.9139	0.9107	0.9130	0.9109
α=1.0	0.9142	0.9143	0.9144	0.9141	0.9144
CIFAR-100, 2.0 s	0.6925	0.6928	0.6926	0.6926	0.6929
CIFAR-100, 0.1 s	0.6927	0.6927	0.6929	0.6927	0.6928
MNIST, 2.0 s	0.9947	0.9949	0.9951	0.9950	0.9951
MNIST, 0.1 s	0.9947	0.9949	0.9950	0.9949	0.9950

(c) External baselines

	nominal 2.0 s				deadline 0.1 s			
Method	Acc.	Timely	MB	Rank	Acc.	Timely	MB	Rank
SA-FedLoRA-style annealing	.9154 ± .0016	0.75	0.29	5.1	.9153 ± .0016	0.62	0.29	5.0
Oort-style utility, fixed rank	.9150 ± .0020	0.76	0.45	8.0	.9149 ± .0019	0.54	0.45	8.0
Joint A/B, plain averaging	.9152 ± .0016	0.73	0.90	16	.9148 ± .0019	0.41	0.90	16
FedLoRA r=16, FP16 upload	.9153 ± .0027	0.77	0.45	16	.9145 ± .0021	0.65	0.45	16
FedRAS, FP16 upload	.9161 ± .0016	0.77	0.45	16	.9156 ± .0014	0.71	0.36	12.8

(d) Scheduling arms (FedLoRA r=16 reference in (a))

	nominal 2.0 s, ten seeds				stress 0.1 s, five seeds			
Arm	Acc.	Timely	MB	Rank	Acc.	Timely	MB	Rank
Full DP (grid assignment)	.9151 ± .0016	0.82	0.90	16.0	.9138 ± .0011	0.55	0.87	15.4
Same-cohort DP	.9152 ± .0019	0.73	0.90	16.0	.9139 ± .0014	0.51	0.70	12.7
Fair cohort (FedRAS rule)	.9153 ± .0016	0.73	0.90	16.0	.9144 ± .0018	0.51	0.70	13.0

(e) Component ablations (nominal deadline)

Configuration	Accuracy	Min.
Full system (ours)	.9144 ± .0013	—
w/o capacity term	.9145 ± .0014	—
w/o frozen A (joint A/B)	.9144 ± .0015	—
w/o rank mask (random ranks, 0.1 s)	.9145 ± .0016	—
w/o server momentum	.9143 ± .0013	—
frozen A, lr 0.02	.9130 ± .0012	0.913
joint A/B, lr 0.02	.9132 ± .0012	0.912

~~agree~~ 2(d) and Table 3(b) repeat the parity at 0.693 on CIFAR-100 and 0.995 on MNIST. At the nominal deadline the rank rule selects rank 16 for every client of every round, so FedRAS coincides with fixed rank 16 in bytes and timeliness, and in accuracy up to floating-point nondeterminism; this identity, rather than a tuned advantage, is what the nominal regime is meant to show. Fig. ~~6~~ 2(a) shows the corresponding trajectories, stable over the 40 rounds for every rank policy.

Delivered-update norms. The capacity law is anchored in measurement: the trainer logs the mean Frobenius norm of each round's delivered updates, and across the full candidate set the measured growth tracks the $\sqrt{r}$ law of (5) — 2.5× from rank 2 to 16 against 2.8× predicted, and 1.9× from rank 4 to 16 against 2.0× — with the stress series coinciding with the nominal one (Fig. 3). The diminishing returns that (5) prices are a measured property of the delivered updates, not a modeling assumption.

6.3. ~~Deadline stress in the trainer~~

~~Table 6~~

6.2.1. Deadline stress

Delivered-rank adaptation. Table 3(a) tightens the deadline to 0.1 s and 0.05 s~~, with ten seeds at 0.1 s and five seeds at 0.05 s~~. Three effects emerge. First, FedRAS downshifts under pressure. Its delivered mean rank falls from 16.0 at the nominal deadline to 12.7 at 0.1 s and 8.1 at 0.05 s, where the delivery hazard (3) makes rank-16 payloads marginal. Second, downshifting pays. At 0.1 s, FedRAS delivers 0.49 timely updates against 0.41 for fixed rank 16 while transmitting 0.69 MB against 0.90 MB, i.e., 23% fewer bytes; at 0.05 s the gap widens to 0.49 against 0.27 timely deliveries at 51% fewer bytes. Third, full-parameter updates collapse completely. Their timely delivery ratio is 0.00 at both stress deadlines, so not a single ~~one of their 5.29 MB updates~~ full-size update arrives on time in any seed; the reported stress accuracy of this row is exactly the per-seed pretrained backbone (0.914 ± 0.002 across seeds), whose spread explains the nonzero standard deviation.

Why downshifting dominates. Payload grows linearly in the rank while delivery probability falls convexly once transmission time approaches the deadline, so the scheduler buys delivery probability where it is cheap and holds high rank where it is free. In this regime the binding constraint on

learning is delivery, not adaptation capacity, which is why the delivered rank moves while accuracy stays at parity.

Against low-rank baselines. The same cells place the remaining policies on one delivery–capacity trade-off: random-rank, ~~and their bytes are attempted in vain at 58.7× the cost~~Oort-style, and annealed schedules deliver *more* timely updates at *fewer* bytes than FedRAS (0.56, 0.54, and 0.62 versus 0.49; 0.42, 0.45, and 0.29 versus 0.69 MB; Table 3(a),(c)) while holding delivered ranks of 6.2, 8.0, and 5.0 against 12.7. No network-side dominance over them is claimed; the claim is that FedRAS reaches comparable delivery behavior while preserving roughly twice the delivered capacity at equal accuracy, dominates fixed rank 16 on both network metrics within that capacity-preserving regime, and prices the trade-off from observed link state rather than a fixed schedule. Whether preserved capacity converts into accuracy in regimes where capacity matters is tested in Section 6.2.2.

~~Real trainer under deadline stress. Accuracy is the round-40 mean and standard deviation. Timely delivery ratio, bytes per round, and mean delivered rank are averaged over rounds 30 to 39 at real serialized sizes. The 0.1 s columns use ten seeds; the 0.05 s columns use five seeds.~~

Mechanism and replication. Accuracy spreads remain within about two standard deviations, ~~which is consistent with the parityclaim. The~~ consistent with parity; the scheduling metrics are where the methods separate ~~.~~ (Fig. ~~6 shows the ten-seed stress trajectories,and Fig. 6 visualizes the mechanism directly~~2(b),(c)): the delivered rank of FedRAS holds 16 under the nominal deadline ~~,~~and tracks 13 and 8 under the 0.1 s ~~deadline, and tracks 8 under the~~ and 0.05 s ~~deadline~~deadlines, whereas the annealed baseline follows its fixed heat–cool path regardless of link state. ~~A ten-seed replication confirms these differences~~ All headline differences replicate across the ten paired seeds at conventional significance levels, with ~~full statistics in Section 6. Under the 0.1 s deadline, FedRAS transmits 23% fewer bytes than fixed r=16, with a seed-mean bootstrap 95% confidence interval from −26.6% to −20.6% and a two-sided Wilcoxon p of 0.002; it delivers a higher timely delivery ratio in ten of ten seed pairs, by 22% on average (p=0.002); and its accuracy is statistically equivalent within the pre-registered ±0.5 percentage-point margin (TOST $p<10^{-6}$~~the statistics fixed in advance (Section 6).

~~Delivered mean rank per round for FedRAS under three deadlines (ten seeds for 2.0 s and 0.1 s, five seeds for 0.05 s), against the fixed annealing schedule of the SA-FedLoRA-style baseline at 0.1 s. FedRAS is link-aware,~~

holding high rank when possible and downshifting under pressure.

Real-trainer accuracy trajectories under the 0.1 s deadline (mean over ten seeds). All methods remain stable; full-model updates deliver nothing and stay at the backbone.

6.3. *Broader stress matrix with mobility, heterogeneity, and cross-dataset checks*

Table 6

6.2.1. *Extended stress matrix*

Table 3(b) extends the study along further axes. A mobility-only scenario with Beta$(0.7, 5)$ mobility at the nominal deadline leaves all LoRA policies at parity, so mobility alone does not force downshifting at this deadline. Extreme label heterogeneity with α=0.1 reveals a second-order effect: rank-16 updates lose 0.32 percentage points relative to rank 4(0.9107 versus 0.9139), consistent with capacity-amplified client drift, and full-parameter updates lose the most(0.9022). FedRAS follows the rank-16 trajectory here, so a heterogeneity-aware downshift is a natural extension. Loosening the deadline to 10 s restores full-model timely delivery to 0.52 without changing any LoRA accuracy. Repeating the 0.1 s deadline on CIFAR-100 and MNIST reproduces the CIFAR-10 stress picture. The scheduling conclusions are therefore not dataset-specific.; Fig. 4 shows the matching sensitivity sweeps over provisioned width, cohort size, and α. Per-decision scheduling cost is negligible in all these settings: the per-client rank rule costs sub-millisecond per decision, orders of magnitude below the full grid-scheduling DP of Section 6.3.1.

6.3. *External baselines*

Table 6 adds three external baselines to the trainer. The first is an SA-FedLoRA-style schedule that heats the rank from 2 to **SAGIN link profile.** Grouping the 20 clients into ground, aerial, and satellite classes with distinct rate and mobility laws reproduces the aggregate parity at the nominal deadline while exposing the expected per-class structure: timely delivery is 0.71 for ground, 0.61 for aerial, and 0.49 for satellite clients. At this deadline the scheduler holds rank 16 and cools it down over the 40 rounds, oblivious to link state [19]. The second is an Oort-style selector that samples $2K$ candidates and admits the top-K by a data-size-and-recency utility at fixed rank [31]. The third is joint A/B training with plain averaging, the

aggregation-level competitor of Section 4.2 [24]across all classes, so the class gap reflects the link laws alone rather than rank allocation. This profile is a parameterized link model, not a measured trace; its role is to show that the scheduler responds sensibly to client-class heterogeneity.

External baselines in the real trainer on non-IID CIFAR-10 with ten seeds. Accuracy is measured at round 40. Timely delivery ratio, bytes, and delivered mean rank are averaged over rounds 30–39.

6.2.1. External baselines

Baseline behavior. Table 3(c) reports the three external baselines of the setup in the real trainer. The link-oblivious annealing schedule settles at low rank and therefore buys timely delivery with only 0.29 MB per roundat the lowest byte cost among the baselines, but it cannot increase rank when links improve, and its capacity ceiling is rank 5. . The Oort-style selector improves selection at fixed rank. Joint A/B training matches frozen-factor training at the adopted learning rate, and the ablations find no stability separation at an elevated learning rate either, so the case for the frozen factor rests on exactness and upload reduction rather than on an observed stability gap.

6.3. Full grid scheduling, same-cohort scheduling, and fair cohort

Compression baseline. A fixed rank-16 policy with an FP16 upload (two bytes per parameter, real round-trip quantization on delivery) halves the payload and, at the 0.1 s deadline, delivers 0.65 timely updates at 0.45 MB — more timely updates at fewer bytes than any FP32 schedule (Table 3(c)). Stacking FedRAS on the same format extends the gain: FedRAS-FP16 beats FP16 rank 16 in five of five seed pairs (+9% timely, −21% bytes, p=0.062 each, the minimum possible at five seeds) at equal accuracy. Compression re-prices every rank; the scheduler re-optimizes the assignment under the prices that hold.

6.2.1. Scheduling variants

The trainer results so far use fair-cohort scheduling: the server samples the fixed cohort of ten clients and assigns eachcandidate its locally optimal rank. **Arms under evaluation.** To attribute the value of exact scheduling, we evaluate two additional arms the three arms defined in the setup are compared in the real trainer, over ten seeds at the nominal deadline and five seeds under stress. The full grid arm solves the dynamic program of Section 4 GridDP assignment (Section 4.1) over the entire 20-client population under

the participation cap and a 1.0 MB budget, which equals ten rank-16 grid costs, so the budget binds only through downshifting or skipping. The same-cohort arm draws the identical cohort as fair cohort and restricts the same dynamic program to those candidates~~, isolating in-cohort rank allocation from candidate-pool widening~~.

~~Scheduling arms in the real trainer. Top: nominal 2.0 s deadline over ten seeds. Bottom: 0.1 s deadline over five seeds. Both arms of each comparison share the candidate protocol described in the text.~~

~~Table 6~~ **Where exactness pays.** Table 3(d) yields a clean decomposition. At the nominal deadline, the same-cohort DP coincides exactly with the per-client argmax rule, as Proposition 2 predicts when the all-rank-16 profile fits the budget; the ~~entire +9 percentage-point~~ full DP's timely-delivery ~~gain of the full DP(0.82 against 0.73) comes~~ gain (+9 percentage points) comes entirely from admission over the wider candidate pool. Under the 0.1 s deadline the pattern persists: the in-cohort DP matches the argmax rule, while the full DP ~~reaches 0.55 against 0.51 timely deliveries at 0.87 against 0.70 MB~~delivers more timely updates at proportionally more bytes. Exactness therefore pays through admission control over a heterogeneous pool, not through rank allocation inside a cohort, where the cheap per-client rule is already optimal (Proposition 2). This delimits when full grid scheduling is worth its coordination cost.

6.3. ~~Scheduler behavior at 500-client scale~~

~~Table 6 summarizes the simulatoracross regimes, with five seeds at the final round. The accuracy column is~~

6.2.1. Component ablations

Component contributions. Table 3(e) reports component ablations at the nominal deadline. In the parity regime, no removal measurably changes final accuracy, which is consistent with the claim that accuracy is carried by the backbone and not by the scheduler. The components instead shape scheduling behavior and aggregation correctness. Removing the capacity term collapses FedRAS to the minimum rank, the degenerate delivery-only policy that the capacity term exists to prevent; its capacity cost is invisible at nominal deadlines but material in the simulator, where reliability-only optimization loses 28 points of utility against rank 16 (Table 4(a)). Removing server momentum is accuracy-neutral here. The rank mask itself is network-metric-neutral at fixed rank policy: with the

~~bounded utility state of Section 6, not a dataset accuracy~~ random-rank rule at the 0.1 s deadline, masked and unmasked aggregation give 0.9143 and 0.9145 final accuracy with identical byte and timeliness profiles. The mask is nonetheless what keeps heterogeneous-rank aggregation per-slice exact whenever ranks mix, as Proposition 3 states, and its absence dilutes high-rank coordinates under mixed-rank rounds.

Stability at elevated learning rate. Under the elevated learning rate of 0.02, the frozen-factor rule and joint A/B training are statistically indistinguishable, with final accuracies of 0.9130 and 0.9132 and transient minima of 0.913 and 0.912 across rounds and seeds. The stability separation between the two aggregation rules observed in our preliminary stylized-payload experiments does not survive the corrected implementation at this horizon, so the frozen-factor rule is justified by the exactness of Proposition 3 and by its single-factor uploads, not by an empirical stability margin at this scale.

6.2.2. Weak-backbone capacity payoff

When capacity becomes valuable. The strong-backbone regime makes adaptation capacity free, so preserving rank carries no measurable benefit (Section 6.2). Weakening the backbone to a one-epoch pretrain (backbone accuracy 0.530) breaks that regime: federated adaptation now improves on the pretrained weights by roughly three and a half points, and the rank–accuracy coupling appears at the nominal deadline (Fig. 5a), where rank 16 reaches 0.564 against rank 4's 0.543 over five paired seeds (+2.1 points, Wilcoxon p=0.062, the minimum possible at five seeds), with FedRAS coinciding with rank 16. Under the 0.1 s deadline the coupling persists (+2.7 points for rank 16 over rank 4) and the scheduler prices it without any retuning (Fig. 5b): FedRAS holds a delivered rank of 12.8, finishes 1.4 points above rank 4 (five of five pairs), and gives back 1.4 points to fixed rank 16 in exchange for a 13% higher timely-delivery ratio and 20% fewer bytes. Preserved capacity therefore converts into accuracy exactly when the backbone is weak enough to need adaptation, which is the regime in which a rank scheduler has a reason to exist.

6.3. Large-scale scheduling simulation

6.3.1. Scheduler behavior across operating regimes

Regime comparison. Table 4(a) summarizes the simulator across regimes at the final round. In the nominal regime, FedRAS matches fixed rank 16 ~~at 0.914~~ and dominates every low-rank policy. Under the 0.1 s

deadline, FedRAS ~~reaches 0.781 against 0.763 for~~ leads fixed rank 16 at 21% fewer bytes, and full-model updates deliver nothing. Under the binding 1.2 MB budget, full-model updates cannot be admitted even once, while FedRAS and rank 16 tie~~at 0.797~~. The mobility-stress and 1000-client regimes show the same pattern: FedRAS and fixed rank 16 coincide at utility 0.94 with timely delivery 0.85 (five seeds) under mobility stress, and at utility 0.92 with timely delivery 0.73 at 1000 clients, so scheduling separates only where deadlines or budgets bind. A final cell repeats the stylized protocol of the earlier literature: with the $s(r)$ payload model the ordering is preserved~~(FedRAS 0.413, rank 16 0.404, annealing 0.374)~~, confirming that the conclusions are not an artifact of the payload accounting.

~~Scheduler simulation with 500 clients, 150 rounds, and five seeds. The accuracy column is a bounded scheduling proxy, not dataset accuracy. Timely delivery ratio is the mean delivery probability for admitted clients, and MB is the attempted uplink payload total for admitted clients in the final round. FedAvg refers to full-model updates at their real 5.29 MB size.~~ **Where the simulator gain comes from.** Fixed rank 16 spends uniformly — weak-link clients miss deadlines while strong-link clients are under-served — whereas FedRAS reallocates, holding high ranks where delivery survives them and downshifting or skipping the rest. Reliability-only optimization fails the other way, buying timely delivery with capacity. The dynamic program between these extremes prices capacity and delivery in a common unit, and the grid makes the resulting knapsack exactly solvable.

Trajectories. Fig. 6 shows the nominal and 0.1 s trajectories, and Fig. 7 the network-side view across regimes. Across the reported regimes, FedRAS is numerically best or tied for best ~~; these simulator comparisons are descriptive unless an accompanying statistical test is reported~~in the utility column of Table 4(a), with the reliability-only and annealed policies retaining a timely-delivery lead bought at substantially lower delivered capacity.

6.4. ~~Solver optimality and runtime~~

6.3.1. Solver optimality and runtime

~~Table 6~~ **Heuristic gaps and runtime.** Table 4(b) reports the scheduling study with the identical ceiling discretization for the DP and every heuristic. When the byte budget is slack, value greedy ~~is optimal~~ and the fixed rank-16 rule ~~is optimalwhenever rank 16 is feasible for all cap clients~~are already optimal, so the choice of scheduler matters little; the value-per-byte greedy, by contrast, collapses to a smallest-rank preference and already loses up

Table 4: Experiments at 500-client scale (five seeds unless noted). (a) Scheduler simulation across regimes; the utility column is the bounded scheduling utility of Section 6 and must not be read as dataset accuracy. MB is the attempted uplink payload of admitted clients in the final round billed on the same 0.01 MB ceiling grid as budget admission, and FedAvg refers to full-model updates at their real serialized size. (b) Solver study with K=25 and 30 instances per cell on the identical ceiling grid; the table reports heuristic objective loss relative to the grid-based DP and DP runtime.

(a) Scheduler simulation across regimes

	Nominal 2.0 s			Deadline 0.1 s			Budget 1.2 MB		
Method	Util.	Timely	MB	Util.	Timely	MB	Util.	Timely	MB
FedAvg (full)	.310	.34	26.5	.093	.00	26.5	.093	.00	0.0
r=4	.744	.75	0.8	.689	.64	0.8	.744	.75	0.8
r=16	.914	.73	2.5	.763	.42	2.5	.797	.76	1.2
Random	.816	.75	1.3	.709	.57	1.3	.803	.75	1.2
Reliability-only	.635	.76	0.5	.606	.70	0.5	.635	.76	0.5
SA-annealing	.842	.76	0.5	.730	.70	0.5	.796	.76	0.5
FedRAS (ours)	.914	.73	2.5	**.781**	.50	2.0	**.797**	.76	1.2

(b) Solver study

Deadline (s)	Budget (MB)	Greedy val/byte	Greedy value	Cohort myopic	Fixed r=16	DP time (ms)
2.0	30	63.6%	0.0%	40.9%	0.0%	421
2.0	4	63.6%	0.0%	40.9%	0.0%	51
2.0	1.2	47.8%	28.4%	48.9%	28.4%	23
1.0	30	62.7%	0.0%	40.0%	0.0%	415
1.0	4	62.7%	0.0%	43.0%	0.0%	51
1.0	1.2	47.3%	29.3%	48.3%	29.3%	23
0.1	30	52.8%	0.0%	51.9%	0.0%	415
0.1	4	52.6%	0.0%	52.0%	0.0%	51
0.1	1.2	41.0%	36.0%	54.8%	36.0%	23

to 6463.6% of the objective in the slack cells. The interesting regimes are the binding ones. When : once the budget bindsat 1.2 MB, value-per-byte greedy loses 41 to 48%, value greedy 28 to 36% , and , every heuristic loses a substantial fraction of the objective — up to 54.8% for the cohort-myopic rule 48 to 55%. The — while the DP reaches the optimum of the discretized single-round estimated problem in every cell. Its runtime is 23 to 421 ms sub-second for 500 clients depending on the budget-grid size, and it across the evaluated grid widths and scales linearly to 1000 clientsat 851 ms, as Fig. 8 shows. This , which is compatible with per-round scheduling at minute-scale FL rounds.

A dedicated grid sweep with steps from 0.02 to 0.005 MB compares each step against a 0.0025 MB reference. In the binding 1.2 MB regime the default 0.01 MB step deviates by at most 3.1%, and a 0.005 MB **Why the heuristics fail where they do.** Value-per-byte pricing fails structurally: the capacity gain $G(r)$ saturates while payload grows linearly, so its ratio always favors the smallest rank regardless of link state. Fixed rank 16 and value greedy fail only under scarcity, where choosing which candidates to drop is the coupling that per-client rules cannot see. The DP runtime follows the budget grid width rather than the client count, which is why it is fastest under exactly the binding budgets that need it most.

Grid sweep. A dedicated grid sweep around the default step bounds the deviation to 2.4% ; slack regimes stay quantization error at about 3% in the binding regime and below 1% in slack regimes; a finer step reduces it further at a known runtime cost. The grid exactness claim of Section 4 5 therefore applies to a grid whose quantization error is bounded, measured, and adjustableat a known runtime cost.

Deployment recipe. The results translate into three planning rules for a practitioner. Calibrate the deadline ladder from the real payload of the widest provisioned rank, so that the nominal deadline leaves rank $r_{\max}$ near-constraint-free while at least one stress deadline drives its timely delivery below one half. Size the byte budget at ten times the grid cost of rank $r_{\max}$, which forces the budget to bind only through downshifting or skipping. Finally, by Proposition 2, use the per-client argmax rule when the budget accommodates every candidate at rank $r_{\max}$ and invoke the full dynamic program only when the pool is heterogeneous enough that admission control matters.

Solver study with 500 clients, K=25, and 30 instances per cell, all methods on the identical ceiling grid. The table reports heuristic objective

loss relative to the grid-based DP and DP runtime.

6.3.2. Delivery-predictor sensitivity

6.4. Sensitivity, link profile, and predictor calibration

Fig. 4 varies the provisioned adapter width, the cohort size, and the label heterogeneity of the trainer on CIFAR-10. Each variant uses five seeds; the nominal reference column uses ten. Widening the adapter from $r_{\max} = 8$ to 32 leaves accuracy flat while bytes rise from 0.45 to 1.80 MB and timely delivery falls from 0.75 to 0.70; the provisioned width is a network-facing decision rather than an accuracy control. Cohort size from 5 to 20 leaves accuracy flat at linearly scaled bytes. Data heterogeneity from $\alpha = 0.1$ to 1.0 moves accuracy as Table 6 shows at constant network cost. Per-decision scheduling cost is negligible in all settings: the per-client rank rule costs about 0.1 ms per decision, roughly 1 ms per round, against 23 to 421 ms for the full grid-scheduling DP of Section 6.3.1.

SAGIN link profile. Grouping the 20 clients into ground, aerial, and satellite classes with distinct rate and mobility laws reproduces the aggregate parity at the nominal deadline (accuracy 0.9141) while exposing the expected per-class structure: timely delivery is 0.71 for ground, 0.61 for aerial, and 0.49 for satellite clients. At this deadline the scheduler holds rank 16 across all classes, so the class gap reflects the link laws alone rather than rank allocation. This profile is a parameterized link model, not a measured trace; its role is to show that the scheduler responds sensibly to client-class heterogeneity, and calibration on measured traces remains open.

Robustness to predictor error. The calibration study of Fig. 9 doubles as a robustness probe. Injecting multiplicative lognormal noise with $\sigma = 0.2$ into the predicted $\hat{P}_{\text{on}}$ yields expected calibration error up to 0.078 while the assignment objective degrades by at most 0.95%; no cell exceeds 1%. Mis-prediction therefore costs optimality smoothly rather than catastrophically, and the observed delivery outcomes of Algorithm 6 are exactly the signal from which a calibrated predictor would re-fit.

Sensitivity of the trainer to provisioned adapter width, cohort size, and label heterogeneity on CIFAR-10 with FedRAS (five seeds per variant; the nominal reference column uses ten seeds). Accuracy is flat in $r_{\max}$, cohort size, and α across the tested ranges; their effect is on bytes and delivered updates (red, right axis).

Delivery-predictor calibration study (500 clients, 150 round-instances per cell). The left panel shows reliability at the 0.1 s deadline. With oracle

link state the hazard-form predictor is calibrated by construction, while the power-distorted (γ) and noisy variants are measurably miscalibrated. The right panel shows the assignment-objective loss of optimizing against the distorted predictor, evaluated under true probabilities, at most 1% in every cell.

Calibration of the delivery predictor. Calibration itself is measurable, and its scheduling cost is bounded, as Fig. 9 shows. On the link model of Section 3, the measures both the fidelity of the deployed predictor with oracle link state and the scheduling cost of its errors (500 clients). With the oracle link state of the setup, the hazard-form predictor (3) is calibrated by construction, with an expected calibration error of 0.003 to 0.004. Deliberately mis-calibrated variants $\hat{P}^{\gamma}$ with γ of 0.5 and 2 reach an error up to 0.22. The . Deliberately distorted variants — power distortions of the predicted probability and multiplicative lognormal noise — are measurably miscalibrated, yet the resulting assignment-objective loss, evaluated under the true probabilities, stays below 1% in every cell. Its placement The loss follows the structure of the predictor's error: the γ-distorted predictors cost more error: power distortions cost most under the stress deadline(up to 0.45%), where rank ordering matters most, while predictor noise costs the noise costs most in the nominal slack cell(0.95%), where it distorts perturbs an otherwise near-uniform rank preference. Even sizeable miscalibration therefore leaves the rank ordering largely intactMis-prediction therefore degrades the objective smoothly rather than catastrophically, and the observed outcomes expose the miscalibration and support delivery outcomes of Algorithm 1 are exactly the signal from which a per-client recalibrationpredictor would be recalibrated. Calibration on measured traces remains open and is taken up in Section 7.

6.4. Component ablations

Table 6 reports component ablations at the nominal deadline. In the parity regime, no removal measurably changes final accuracy, which is consistent with the claim that accuracy is carried by the backbone and not by the scheduler. The components instead shape scheduling behavior and aggregation correctness. Removing the capacity term collapses FedRAS to the minimum rank, the degenerate delivery-only policy that the capacity term exists to prevent; its capacity cost is invisible at nominal deadlines but material in the simulator, where reliability-only optimization loses 28 points of utility against rank 16 (Table 6). Removing server momentum

~~is accuracy-neutral here. The rank mask itself is network-metric-neutral at fixed rank policy: with the random-rank rule at the 0.1 s deadline, masked and unmasked aggregation give 0.9143 and 0.9145 final accuracy with identical byte and timeliness profiles. The mask is nonetheless what keeps heterogeneous-rank aggregation per-slice exact whenever ranks mix, as Proposition 3 states, and its absence dilutes high-rank coordinates under mixed-rank rounds.~~

~~Under the elevated learning rate of 0.02, the frozen-factor rule and joint A/B training are statistically indistinguishable, with final accuracies of 0.9130 and 0.9132 and transient minima of 0.913 and 0.912 across rounds and seeds. The stability separation between the two aggregation rules observed in our preliminary stylized-payload experiments does not survive the corrected implementation at this horizon, so the frozen-factor rule is justified by the exactness of Proposition 3 and by its single-factor uploads, not by an empirical stability margin at this scale.~~

~~Component ablations, real trainer (non-IID CIFAR-10, five seeds, nominal deadline). The Min. column is the minimum over rounds of the per-round seed-mean accuracy in the elevated-learning-rate stability check at learning rate 0.02.~~

6.4. ~~Summary of findings~~

~~Across both tiers , rank~~ **Cross-tier summary.** Taken together, the two evaluation tiers give a consistent picture. Rank choice does not measurably change accuracy for the frozen backbone ~~and~~ at the 40-round horizon, and every accuracy comparison against fixed rank 16 passes the ~~pre-registered~~ TOST equivalence test fixed in advance at the ± 0.5 percentage-point margin~~. The main differences are on the network side. Full-parameter updates cost 58.7× the bytes of a rank-16 upload, miss 83% of nominal deadlines and all stress deadlines, and fail to improve on the backbone when only sparse updates arrive~~; an offline control ~~without delivery modeling trains~~ that ignores delivery modeling entirely and trains on every scheduled update reaches 0.913 ± 0.002 at the nominal deadline, close to the backbone~~level~~'s 0.914, separating learning capacity from delivery feasibility.

~~FedRAS behaves as designed, holding~~ The methods instead separate on the network side: FedRAS holds high ranks on healthy links and ~~downshifting under pressure. Over ten seeds at the 0.1 s deadline it transmits 23% fewer bytes than fixed rank 16 (bootstrap 95% CI [−26.6%, −20.6%], Wilcoxon p=0.002, matched-pairs rank-biserial 1.0), delivers 22% more timely updates~~

~~(p=0.002), and is accuracy-equivalent. At the nominal deadline it reduces to fixed rank-16 behavior at zero cost. The exact dynamic program~~ downshifts under pressure, full-parameter updates collapse once deadlines bind at real payload sizes, and the exact scheduler pays through admission over the ~~wider candidate pool , +9 percentage points of timely delivery at the nominal deadline and +8% under stress , while the per-client argmax rule is already optimal inside a cohort (Proposition 2~~candidate pool rather than in-cohort rank allocation (Proposition 2). Two stress tests bound these claims from both sides: the network-side gains survive stacking on an FP16 codec (Section 6.2.1), and a weaker backbone reveals accuracy value in the preserved capacity exactly where the strong-backbone regime shows none (Section 6.2.2). These tests rest on ten-seed replications of the headline trainer regimes~~with common random numbers within each seed~~; they support the stated network-side comparisons and the accuracy equivalence, and they do not establish accuracy dominance. Section 7 interprets these findings, states their scope conditions, and translates them into deployment guidance.

7. Discussion and limitations

7.1. Interpretation and deployment considerations

~~Scope of claims~~Why the gains are network-side. The trainer demonstrates parity and equivalence, not rank-dependent accuracy gains. With a strong frozen backbone and 40 adaptation rounds, transmitted rank does not measurably change test accuracy, because the accuracy is carried by the pretrained features while the scheduler shapes which updates arrive on time and at what byte cost. Claims that low rank limits adaptation quality would require longer horizons, weaker backbones, or distribution-shifted clients.

When FedRAS does not help. Three settings bound the method's value. First, when the candidate pool fits within the participation cap and the budget accommodates every candidate at the provisioned width, FedRAS reduces exactly to fixed rank 16 (Proposition 2): the scheduling layer adds coordination cost and nothing else. FedRAS is therefore worth its coordination cost when deadlines are tight relative to real payloads or when the byte budget binds; outside those regimes, fixed rank 16 is the identical and simpler choice. Second, under extreme label heterogeneity, higher ranks lose accuracy relative to low ranks through capacity-amplified

client drift, and FedRAS follows the high-rank trajectory and inherits that loss; conditioning the capacity term $G(r)$ on measured heterogeneity is left as future work. Third, under tight deadlines the weakest client class still delivers least, because downshifting protects delivery but cannot create link capacity.

Deployment recipe. The results translate into three planning rules for a practitioner. Calibrate the deadline ladder from the real payload of the widest provisioned rank, so that the nominal deadline leaves rank $r_{\max}$ near-constraint-free while at least one stress deadline drives its timely delivery below one half. Size the byte budget at ten times the grid cost of rank $r_{\max}$, which forces the budget to bind only through downshifting or skipping. Finally, by Proposition 2, use the per-client argmax rule when the pool fits within the participation cap and the budget accommodates every candidate at rank $r_{\max}$; invoke the full dynamic program whenever the pool exceeds the cap or the budget forces admissions to compete.

Coordination cost and integration. FedRAS is an aggregator-side module and leaves client training untouched: clients receive their assigned rank with the round schedule, slice their factors to it, and run the local objective as before. The scheduler's inputs are the estimated uplink rates, the link-state history, and the payload table of the provisioned width; its outputs are the admission and rank decisions attached to the schedule. The aggregation change is confined to the rank mask of Section 4.2, a masked average rather than a new protocol. Its failure modes are bounded: a degraded predictor degrades the objective smoothly (Fig. 9) rather than derailing the schedule, and if link-state estimates are unavailable altogether the scheduler falls back to fixed $r_{\max}$, which Proposition 2 identifies as optimal whenever the pool fits the cap and the budget accommodates every candidate at that rank.

7.2. Scope and limitations

Scope of claims. The results describe network-side gains in the evaluated frozen-backbone setting; they must not be read as universal accuracy gains. The simulator capacity model is a scheduling abstraction, and its accuracy column is a bounded scheduling proxy rather than a measured dataset accuracy.

Backbone pretraining. The centralized pretraining stage is a documented premise inherited from parameter-efficient FL practice; the federated phase adapts but does not train the backbone. Where no server-side corpus

exists, self-supervised or public-data pretraining would be needed, and the parity result should be re-verified.

Estimator assumptions. The hazard-form delivery model (3) assumes the closed-form exponential realization, and the experiments supply generated link states as oracle estimates; training and processing times are included in the conceptual delivery model (1) but are not separately measured. Production deployments should calibrate P_{on} per client from observed outcomes, for example with isotonic or Platt scaling monitored by a Brier score, and the scheduler treats the predictor as a plug-in. Calibration on measured wireless traces remains open.

Myopia. The ~~DP is exact for the estimated~~ analysis is deliberately per-round ~~problem, not for the multi-round stochastic control problem induced by budget carry-over and mobility correlation~~(Section 5); a horizon-aware policy ~~could shift~~ that shifts admissions across rounds ~~and~~ as budgets carry over and mobility correlates is a natural extension.

~~Security~~Evaluation horizon and topology. ~~This work schedules communication and does not propose protection, so we make no privacy claim. Frozen-factor training incidentally reduces the dimensionality of transmitted gradients, which shrinks the gradient-inversion surface. Formal statements require differential-privacy accounting or secure aggregation, which compose orthogonally with rank scheduling [40, 41].~~ The trainer conclusions are horizon-bound: forty adaptation rounds on a strong frozen backbone is precisely the regime in which rank choice is accuracy-neutral, and longer horizons, weaker backbones, or distribution-shifted clients could re-open an accuracy gap that the equivalence tests employed here cannot see. The evaluation also assumes a star topology with a trusted aggregator that collects estimated link states and computes the schedule. Nothing in the formulation requires clients to report state beyond the delivery outcomes any FL round makes observable, but a decentralized or aggregator-less variant would need a different synchronization point and is not studied here.

~~When FedRAS does not help.~~ ~~Three settings bound the method's value. First, when links are healthy and the budget accommodates every candidate at the provisioned width, FedRAS reduces exactly to fixed rank 16 (Proposition 2): the scheduling layer adds coordination cost and nothing else, so the simpler fixed-rank configuration is preferable. Second, under extreme label heterogeneity (α=0.1), higher ranks lose 0.32 percentage points of accuracy relative to rank 4 through capacity-amplified client drift, and FedRAS follows the rank-16 trajectory and inherits that loss; conditioning~~

~~the capacity term $G(r)$ on measured heterogeneity is left as future work. Third, under tight deadlines the weakest client class still delivers least (satellite 0.49 in the grouped profile), because downshifting protects delivery but cannot create link capacity. FedRAS is therefore worth its coordination cost when deadlines are tight relative to real payloads or when the byte budget binds; outside those regimes, fixed rank 16 is the identical and simpler choice~~**Security.** The adversary and trust assumptions are stated in Section 3.4. This work proposes no protection mechanism and makes no privacy claim; differential-privacy accounting, secure aggregation, and Byzantine robustness compose orthogonally with rank scheduling [40, 41].

Generality. The trainer validates adapters on convolutional and linear layers of a CNN at $r_{\max}$=16 with 20 clients, and the simulator extends the population to 1000, both under synthetic link distributions. Extension to transformer backbones, measured rather than synthetic links, and larger rank sets may change the payload slope, capacity curve, and empirical conclusions. The formulation can be tested in those settings. The study does not measure energy, system overhead beyond scheduler runtime, or client fairness and therefore makes no claim about those properties.

8. Conclusion

This paper reframed Federated LoRA over mobile edge networks as a deadline-aware client–rank scheduling problem in which the LoRA rank is a network-visible resource accounted at its real serialized size. We derived a rank-conditioned delivery model, formulated the per-round assignment as a multiple-choice knapsack under budget and participation caps, solved the grid-discretized assignment with a pseudo-polynomial dynamic program, and stabilized heterogeneous-rank aggregation with frozen-factor rank-masked averaging in which only the trainable factor is ever transmitted.

Experiments with a real federated LoRA trainer and a 500-client scheduler simulation, with deadlines calibrated to real payloads, show that ~~FedRAS~~ rank selection preserves backbone accuracy up to ~~a pre-registered equivalence margin ,~~ an equivalence margin fixed in advance while reshaping the network side of the exchange: under ten-seed deadline stress, FedRAS delivers 22% more timely updates and 23% fewer bytes than fixed rank 16 ~~under ten-seed deadline stress, and exposes the complete collapse of full-parameter updates, which spend 58.7× the bytes and deliver nothing on time under stress. The exact scheduler recovers 28 to 55% expected useful updates over greedy~~

~~heuristics when the budget binds, up to 64% over value-per-byte pricing, at sub-second cost, and its measurable value in the trainer comes from admission over the candidate pool rather than~~ and occupies the high-capacity end of the delivery–capacity trade-off on which link-oblivious low-rank schedules sit at the opposite end, and the exact scheduler earns its cost through admission over heterogeneous candidate pools rather than through in-cohort rank allocation. The claims are conditional on the frozen-backbone setting and synthetic link models: rank scheduling adds value when deadlines bind relative to real payloads or when the byte budget binds, and reduces to fixed-rank operation otherwise. Future work includes horizon-aware stochastic scheduling, calibrated per-client delivery predictors on measured traces, protection mechanisms composed with rank scheduling, and validation on transformer backbones.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors thank the members of the Information Engineering University research group for many useful discussions and for hands-on assistance with the experimental implementation.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

CRediT authorship contribution statement

Wenyu Ma: Methodology, Software, Validation, Investigation, Writing – original draft. **Tao Hu:** Conceptualization, Formal analysis, Supervision, Writing – review and editing. **Xinlei Liu:** Data curation, Investigation, Visualization. All authors approved the submitted version and accept responsibility for the work.

Data availability

The reproducibility package accompanying this submission contains the complete source code (trainer, simulator, solver, and analysis scripts), per-run configuration files, per-seed result tables, and the raw training logs supporting every table and figure in this study; a permanent repository DOI will be supplied at publication.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used AI-assisted tools for language editing and formatting support. The authors reviewed and edited all generated material and take full responsibility for the content of the published article.

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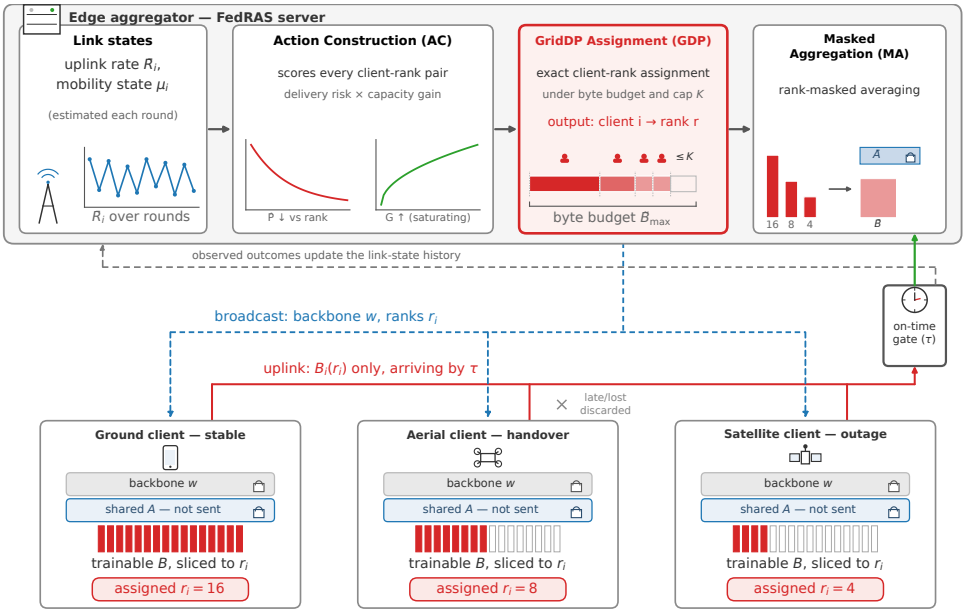
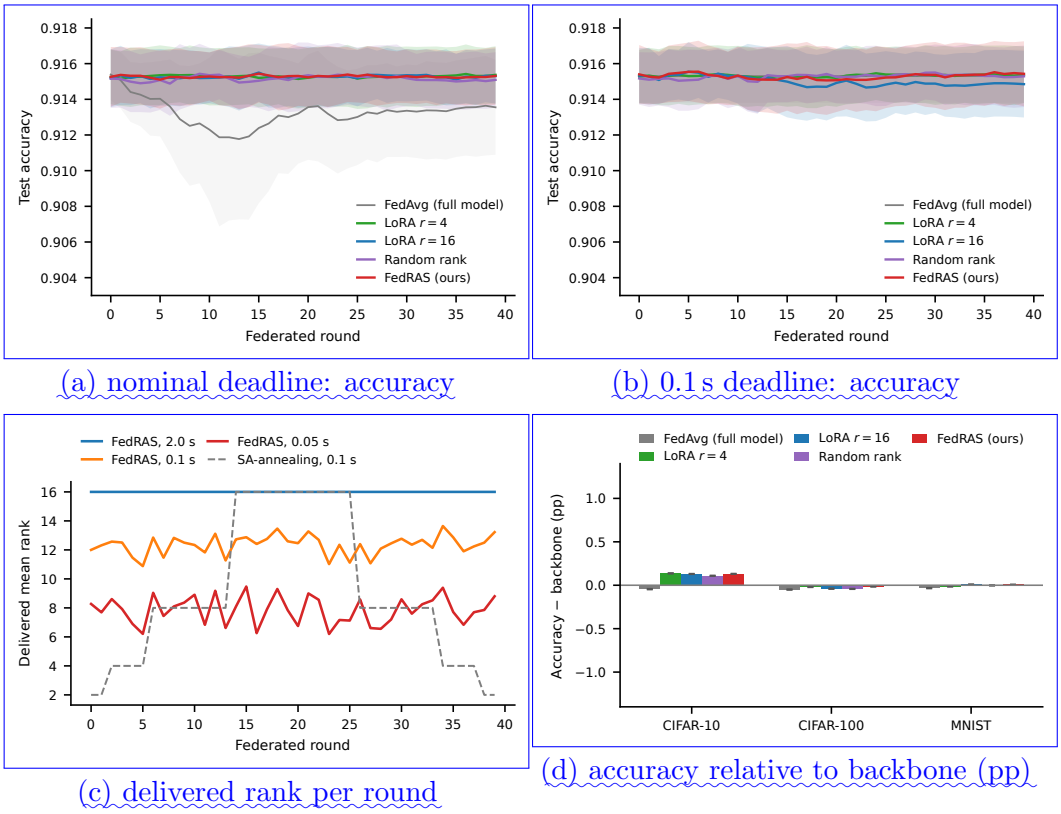


Figure 1: FedRAS ~~system architecture and~~ round pipeline. ~~The edge aggregator estimates client link states,~~ Action Construction (AC) scores every client–rank pair with the delivery hazard (3) and the saturating capacity (5)~~, and~~ ; the GridDP assignment (GDP) computes the client–rank assignment ~~with the exact grid-based dynamic program~~ under the byte budget and the admission cap~~(Proposition 1). Admitted clients train on a frozen backbone with LoRA factors sliced to the assigned rank and upload only the trainable factor B, at 5.63 KB per rank column; the shared factor A is never transmitted~~. Updates arriving within the deadline τ pass the on-time gate and are ~~aggregated rank-maskedly~~ merged by Masked Aggregation (MA) while the shared factor A stays frozen~~(Proposition 3)~~; ~~late or lost updates are discarded, and the~~ observed outcomes update the link-state history. AC, GDP, and MA correspond to Algorithms 2–4; the overall round is Algorithm 1.



(a) nominal deadline: accuracy

(b) 0.1 s deadline: accuracy

(c) delivered rank per round

(d) accuracy relative to backbone (pp)

Figure 2: ~~Final accuracy across datasets.~~ Real-trainer experiments (CIFAR-10~~uses ten~~, means over seeds~~; CIFAR-100~~). (a) At the nominal deadline all rank policies hold the pretrained-backbone accuracy. (b) Under the 0.1 s deadline all methods remain stable while full-model updates deliver nothing and ~~MNIST use five seeds~~stay at the backbone. ~~The parity~~ (c) FedRAS is link-aware: delivered rank holds 16, tracks 13, and ~~equivalence findings~~ tracks 8 under the 2.0, 0.1, and 0.05 s deadlines, against the fixed heat–cool path (gray dashed) that the annealing baseline follows regardless of ~~Table~~link state. (d) ~~6 are~~ Final accuracy relative to the per-dataset pretrained backbone; the parity is not CIFAR-10-specific (numbers in Table 3(b)).

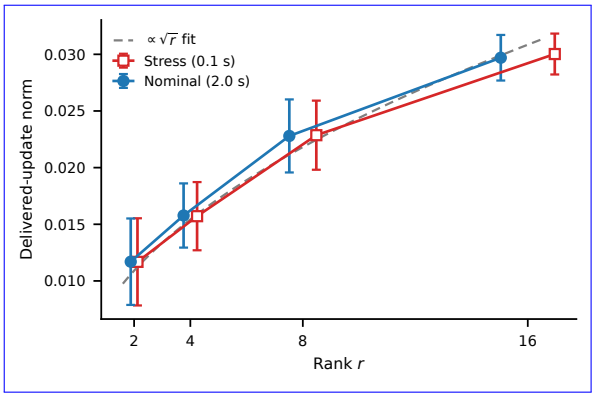


Figure 3: ~~Real-trainer~~ Delivered-update norm vs. rank on CIFAR-10 ~~test accuracy over 40 federated rounds at the nominal deadline~~ (mean over ~~ten~~ delivered rounds; five seeds per point, error bars are one standard deviation). ~~All~~ The measured growth (2.5× from rank ~~policies hold~~ 2 to 16, 1.9× from rank 4 to 16) tracks the ~~pretrained-backbone accuracy~~square-root capacity law of (5) (2.8× and 2.0×) rather than linear payload growth (8× and 4×); ~~differences appear in bytes~~ the nominal and ~~timeliness, not accuracy~~stress series coincide.

~~Extended stress matrix. Results use five seeds and report round-40 accuracy as mean over seeds. The mobility-only and α=1.0 scenarios probe extreme conditions; the last two columns repeat the 0.1 s deadline on CIFAR-100 and MNIST.~~

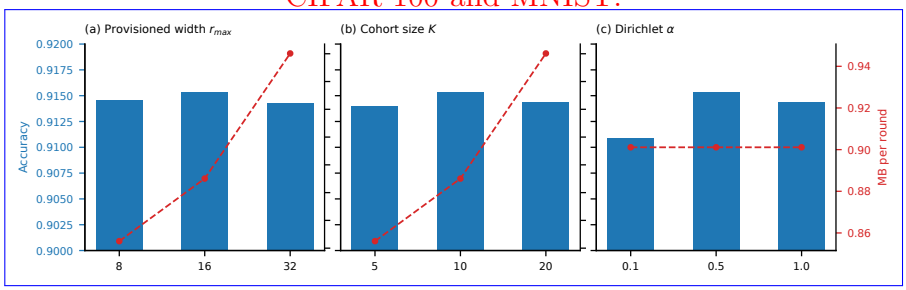


Figure 4: Sensitivity of the trainer to provisioned adapter width, cohort size, and label heterogeneity on CIFAR-10 with FedRAS (five seeds per variant; the nominal reference column uses ten seeds). Accuracy is flat in $r_{\max}$ and cohort size, with a second-order drop only at α=0.1; the dominant effect of all three axes is on bytes and delivered updates (red, right axis).

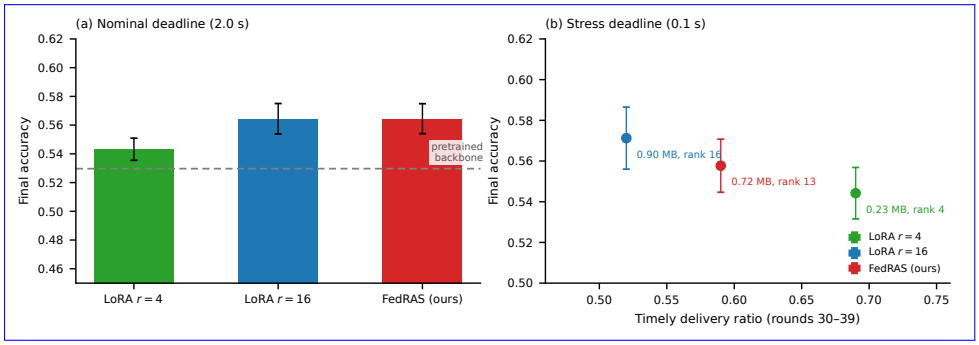


Figure 5: Weak-backbone capacity payoff (one-epoch pretrain, five seeds). (a) Nominal deadline: the rank–accuracy coupling appears (rank 16 above rank 4 by 2.1 points) and FedRAS coincides with rank 16; the dashed line marks the pretrained backbone. (b) Stress deadline (0.1 s): FedRAS lands between the fixed ranks on accuracy while delivering more updates than rank 16 at 20% fewer bytes; annotations give attempted MB per round and mean delivered rank.

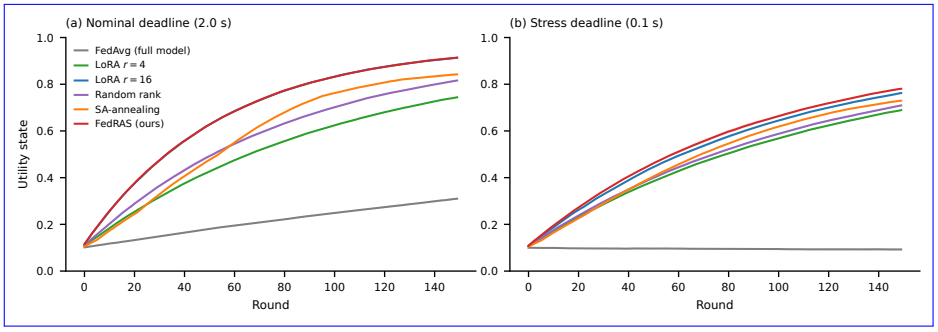


Figure 6: Scheduler simulation utility-state trajectories (mean over five seeds). (a) Nominal 2.0 s deadline, where FedRAS matches fixed rank 16. (b) 0.1 s deadline, where FedRAS downshifting separates from fixed rank 16.

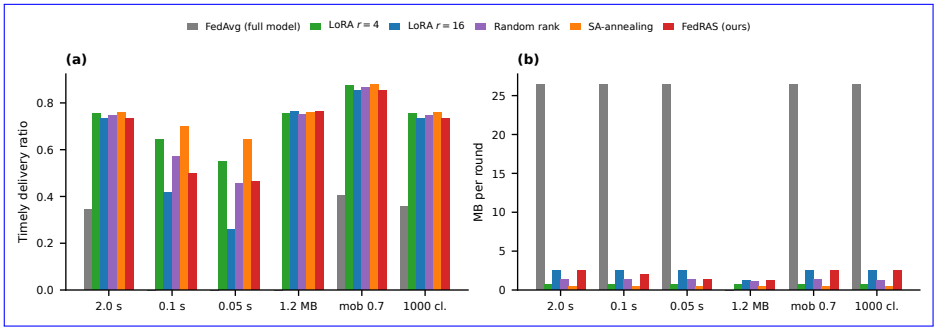


Figure 7: Timely delivery ratio and attempted bytes per method across simulator regimes, including the 1000-client scalability regime (five-seed means, final round). Full-model updates deliver nothing under stress and cannot be admitted at all under the binding budget.

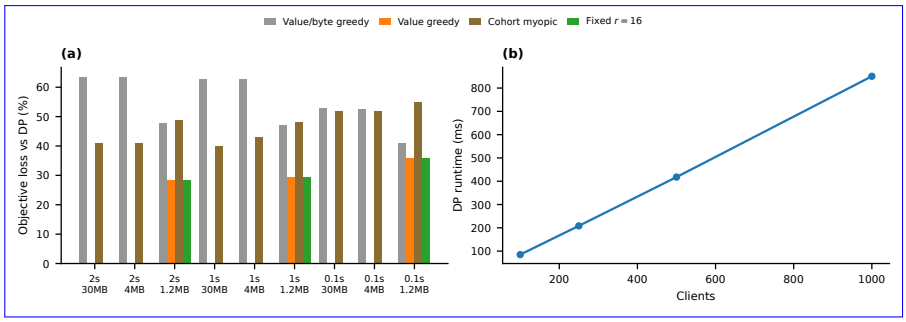


Figure 8: Solver study. (a) Objective loss of the heuristics relative to the DP across deadline and budget cells. (b) DP runtime scaling with the client population.

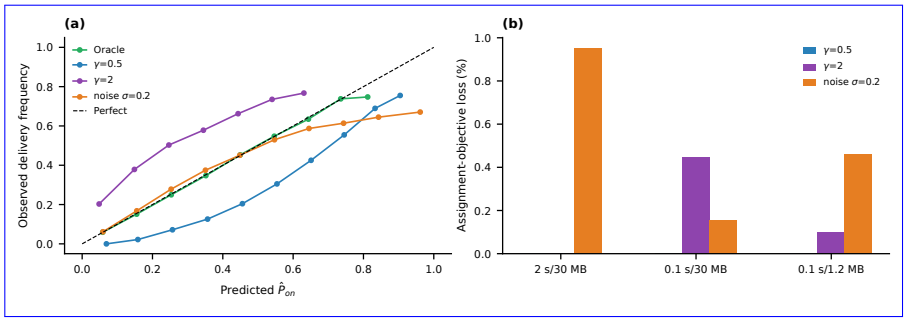


Figure 9: Delivery-predictor calibration study (500 clients, 150 round-instances per cell). Left: reliability at the 0.1 s deadline; with oracle link state the hazard-form predictor is calibrated by construction, while the power-distorted (γ) and noisy variants are measurably miscalibrated. Right: assignment-objective loss of optimizing against the distorted predictor, evaluated under true probabilities, at most 1% in every cell.